# Resolving the Forward Guidance Puzzle: Theory and Evidence from the Great Depression and the Great Recession

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The forward guidance puzzle is usually read as a failure of the New Keynesian model: the model predicts that announcements about future interest rates should move output and inflation sharply, yet post-2008 forward guidance looked modest in the data. We argue the puzzle is a property of the policy regime, not of the model. In the canonical experiment the central bank commits to a fixed interim path and does not respond to inflation or output before the promised date—a regime we call *FG-Passive*. Keep the interim feedback—*FG-Active*—and the same model predicts bounded, modest effects. The two episodes that have anchored the debate fall on opposite sides of this line: post-2008 guidance was *FG-Active*, and its effects were modest; four regime shifts during the Great Depression approximated *FG-Passive*, and output and prices moved by the large magnitudes the model implies. Once the regime is correctly classified, the canonical New Keynesian model is consistent with both episodes.

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## 1. Introduction

Post-2008 forward guidance looked modest in the data. The canonical New Keynesian model says it should have been dramatic. In the exercise documented by Del Negro, Giannoni, and Patterson (2023),[1] a commitment to lower interest rates eight quarters ahead produces an impact several times larger than a cut today. Yet once short-term rates reached the zero lower bound in December 2008 and central banks turned to such announcements as a principal instrument, the effects that followed were modest. That gap is the *forward guidance puzzle*.

It is tempting to read this evidence as a failure of the canonical model. Much of the subsequent literature took the modest effects of post-2008 forward guidance as the empirical benchmark and looked for mechanisms that dampen the predictions of the passive-policy experiment, including work by one of the authors of this paper. The proposed mechanisms include incomplete markets, overlapping generations, departures from rational expectations, and imperfect credibility.[2] This literature is valuable even absent the forward guidance puzzle. It studies how the economy propagates news about future interest rates rates.

The usual response has been to suspect the model. We suspect the question instead. What is forward guidance in the model? In the canonical experiment, forward guidance is not just an announcement about future interest rates. It is a commitment to a future rate path while policy remains passive in the interim. A path is not a rule. A rule tells the central bank what to do if inflation or output changes before the promised date. Sticking to a path tells it not to react. We call this regime *FG-Passive*. The passive path is the key assumption.

There is another regime. The central bank can announce future policy while retaining its normal feedback rule in the interim. If inflation or output rises too much, it raises rates. We call this regime *FG-Active*. Under this regime, the same New Keynesian model gives bounded responses of output and inflation. This was already recognized in the original statement of the puzzle: Del Negro, Giannoni, and Patterson (2023) show that allowing monetary policy to respond to inflation in the interim bounds the response of the economy.[3] We ask which regime describes the historical episodes that have been used to

[1] First released as a New York Fed Staff Report in 2012.

[2] On incomplete markets and heterogeneity, see McKay, Nakamura, and Steinsson (2016); Werning (2015); Bilbiie (2017, 2020, 2025); Caballero and Farhi (2018); Kaplan, Moll, and Violante (2018); Hagedorn et al. (2019); Acharya and Dogra (2020); Michaillat and Saez (2021); and McKay and Wieland (2022). On overlapping generations, see Del Negro, Giannoni, and Patterson (2023) and Eggertsson, Mehrotra, and Robbins (2019). On departures from rational expectations, see Gabaix (2020); Woodford (2019); Chung, Herbst, and Kiley (2014); Kiley (2016); Carlstrom, Fuerst, and Paustian (2015); Angeletos and Lian (2018); Beqiraj, Di Bartolomeo, and Di Pietro (2019); García-Schmidt and Woodford (2019); and Gabaix (2023). On credibility, see Campbell et al. (2019); Bodenstein, Hebden, and Nunes (2012); Haberis, Harrison, and Waldron (2014); Andrade et al. (2019); Bernanke (2020); de Groot and Mazelis (2020); and Lunsford (2020).

[3] Related theoretical points have since been developed by Maliar and Taylor (2024), Bianchi, Melosi, and

judge the model.

Our contribution is two-fold. First, we classify the episodes that have anchored the debate. They fall on opposite sides of the regime line. The Great Recession was active. The Great Depression was passive. This classification explains why the prior literature finds that the canonical model misses during the Great Recession, and why, as we will argue, it doesn't miss during the Great Depression. Accounting for the underlying regime allows the model to match both periods with no heterogeneity, no bounded rationality, no overlapping generations, and no imperfect credibility. To our knowledge, this symmetric reading of the two episodes is new.

Second, we show that the forward-guidance boom is the positive-sign mirror image of the Eggertsson and Woodford (2003) deflationary spiral. Both arise from an inactive short-rate path. They differ only in sign—and in one important economic respect. In the deflationary spiral the central bank is *unable* to respond, because the zero lower bound blocks rate cuts. In the boom it is *able* to respond, by raising rates, and only commits not to. The explosive forward-guidance puzzle can be eliminated by raising rates; the deflationary spiral cannot. Cochrane (2017) studies this connection through equilibrium indeterminacy under an interest-rate peg. We show that it maps onto *FG-Passive* versus *FG-Active*, and that the historical record discriminates between the two.

The mechanism we use is not new, and we do not claim it is. We solve the canonical New Keynesian model under an active interim rule, changing only the policy regime. That an active rule bounds the response is stated in Del Negro, Giannoni, and Patterson (2023). The underlying logic is older still. It is already present in the analysis of the deflationary spiral in Eggertsson and Woodford (2003), of which the forward-guidance puzzle is the mirror image. Our point is what that distinction implies once it is taken to the historical record.

While the Great Recession motivated the puzzle, it is not its empirical counterpart. Post-2008 forward guidance was explicitly state-contingent, conditioned on the inflation and slack outlook, and hedged with escape clauses (Federal Open Market Committee 2012a; Bank of England 2013; European Central Bank 2013). The way central banks communicated forward guidance in the Great Recession is consistent with an active-rule stance whose effects, we argue, should be modest. This is exactly what the empirical literature finds (Campbell et al. 2019; Bundick and Smith 2020).

However, four turning points during the U.S. Great Depression are close counterparts to *FG-Passive*—Roosevelt's 1933 exit from gold and pledge to reflate to the 1926 price level, Eccles's consolidation of the reflation regime in 1934, the "Mistake of 1937," and the reversal of 1938. Each regime change was announced while short rates were near zero and

Nicolò (2024), Eskelinen, Gibbs, and McClung (2024), and Ahn and Holm (2025). In different environments, these papers show that passive or time-dependent pegs generate the large effects associated with the puzzle, while sufficiently state-dependent or active policy rules bound them.

without systematic interim feedback. The inflation objective signalled not just a willingness to accept, but a desire to generate inflation of 10 percent or more per year for the next four years. Such a commitment to accept large, long-lasting inflation without a policy response is much closer in spirit to the *FG-Passive* than anything central banks tried in the Great Recession. Just as the model predicts, the data moved with the regime changes: double-digit swings in output and prices followed the 1933 shift, the 1937 reversal, and the 1938 expansion.

The asymmetry is intuitive: the zero lower bound stops the central bank from cutting, not from raising. A passive path booms because the bank commits not to lean against the boom; any responsive rule would raise rates in the interim and bound the effect. The hedged, state-contingent forward guidance of the post-2008 era signalled a responsive rule. Roosevelts price level objective, on the other hand, was closer to a policy peg. The issue is not that the model is too forward-looking—it is the policy regime.

Our quantitative strategy follows this logic. We calibrate the canonical New Keynesian model to the Depression-era decline and read changes in expected zero-lower-bound duration from movements in the three-year Treasury yield around policy announcements. Feeding these implied horizons into the model yields output and inflation movements similar to those observed in the data. During the Great Recession, on the other hand, State-contingent guidance extended the expected zero-lower-bound duration only modestly and produced correspondingly modest macroeconomic effects, as the model predicts under an active interim rule (Campbell et al. 2019; Bundick and Smith 2020).

Our findings suggest that the forward guidance puzzle is a diagnostic tool for policy regimes rather than a bug of the cannonical model. Large effects of forward guidance, in models or in data, indicate that the interim policy path was inactive. Modest effects indicate a responsive regime with interim feedback. The Depression-era data pass the test for *FG-Passive*. The post-2008 data pass the test for *FG-Active*. The model's predictions are consistent with both episodes once the regime is classified correctly.

These findings do not diminish the value of the extensions developed in the forward guidance puzzle literature. Those models deepen our understanding of propagation mechanisms that matter independently of the policy regime. They may also fit the data better than the rudimentary canonical NK model. Our point is narrower. Regime classification is first-order. Once it is taken into account, the baseline model's predictions are broadly consistent with the historical evidence under both regimes.

The paper proceeds as follows. Section 2 documents the contrasting regimes. Section 3 develops analytics that clarify regime dependence. Section 4 presents the quantitative results for a model calibrated to the Great Depression. The final section concludes with implications for the use and interpretation of forward guidance at the zero lower bound.

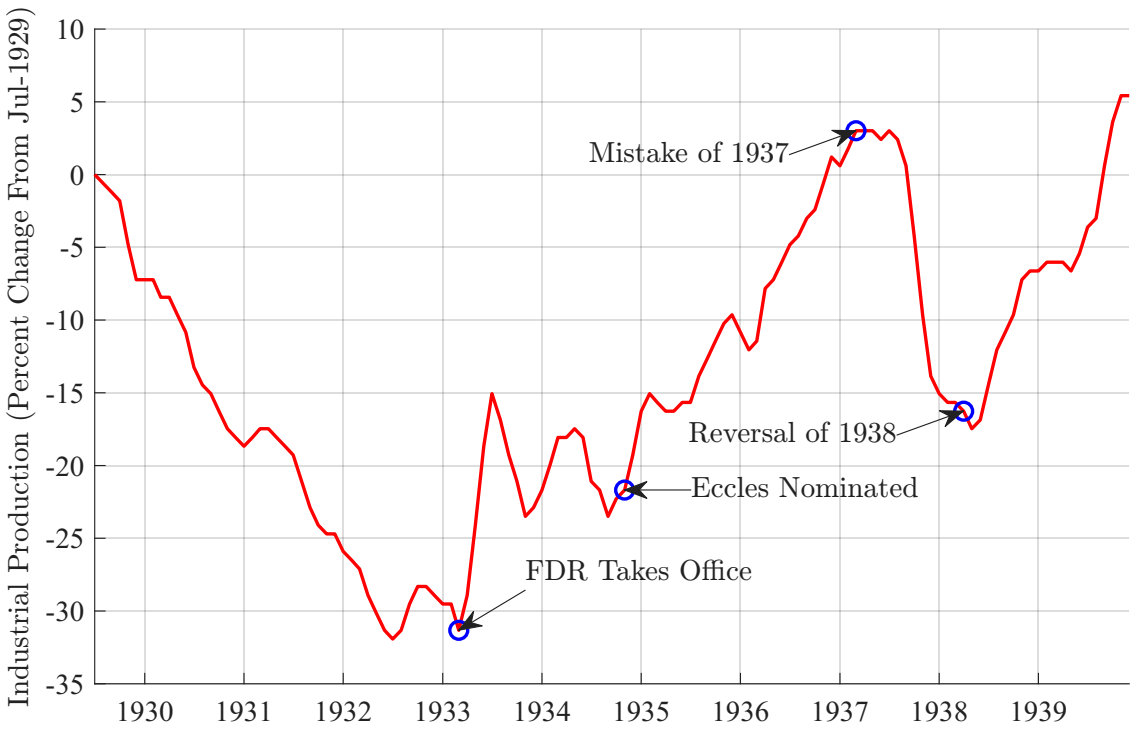



FIGURE 1. Industrial Production During the Great Depression, 1929–1939.

*Note*: The solid red line plots industrial production from January 1929 to April 1939 rescaled so that the June 1932 to March 1933 average matches the 30 percent decline in GDP at the trough of the Depression. The blue markers and labels highlight the dates of the four episodes we discuss below: March 1933 (subsubsection 2.1.1), November 1934 (subsubsection 2.1.2), March 1937 (subsubsection 2.1.3), and April 1938 (subsubsection 2.1.4). We choose to focus on industrial production as it is available monthly and, as Figure A1 in Appendix A shows, the path of industrial production is broadly in line with the path for GNP and GDP. *Source*: FRED.

## 2. Policy Regimes and Historical Evidence

The effects of forward guidance depend crucially on whether the central bank commits to a policy path or continues to adhere to a normal policy rule. Whether a path or a rule better describes policy is an empirical question, so we begin with the historical record. The Great Depression and the Great Recession both featured interest rates near zero, but not under the same regime. The Depression had regime changes: policymakers announced price-level objectives and, for meaningful periods, did not condition them on interim inflation or output. Post-2008 forward guidance was different—explicitly state-contingent, hedged with inflation conditions and escape clauses. The model predicts large effects in the first case and modest effects in the second.

### 2.1. The Great Depression: Four Regime Changes at the Zero Lower Bound

Between 1933 and 1938, U.S. monetary policy changed four times. Each change occurred while short-term interest rates were near zero. Each change was about the future monetary regime. The announcements were not modern threshold guidance. They were not conditioned on a near-term inflation forecast or on an unemployment number. They were commitments about the future price level and the future conduct of policy. Figure 1 marks the four dates, and the swings around them are large.

#### 2.1.1. Episode 1: Roosevelt Takes Office (March 1933)

When Franklin D. Roosevelt took office in March 1933, he broke with the old regime of his predecessor, Herbert Hoover. He abandonned the gold standard and announced a goal: restore prices to their 1926 level (Eggertsson 2008). This was not a small adjustment in the policy rate. Short rates were already near zero. It was a commitment about future monetary conditions. If taken literally, the target implied roughly 10 percent annual inflation for three to four years.

Roosevelt made his thinking clear in a May 1 statement to the *Wall Street Journal*:

> "[O]ur primary need is to insure an increase in the general level of commodity prices. To this end simultaneous actions must be taken in both the economic and the monetary fields.[4]"

This was a sharp break with Hoover. Hoover had defended the gold standard, balanced budgets, and limited government. Roosevelt had also sounded orthodox during the campaign. That made the break more surprising when it came (Jalil and Rua 2016). The financial press did not see it coming. Newspapers during the campaign, and even after the election, expected more continuity with Hoover's policies.

Roosevelt not only spoke; he acted. Government consumption and investment nearly doubled within a year—a deficit-financed program of public works, agricultural support, and direct relief (Eggertsson 2008)—and the administration wrested control over gold and silver purchases from the Federal Reserve, ensuring the monetary base would expand. These were not empty words but institutional changes.

Markets moved with them: commodity prices rose and inflation expectations shifted. Hamilton (1992) shows this using commodity futures. The earlier deflation had largely been anticipated, and was expected to continue, so the 1933 change was a genuine break (Cecchetti 1992). Jalil and Rua (2016) document a sharp rise in newspaper discussions of inflation after Roosevelt left gold on April 19. By late May, business analysts in the *Economist*, *Business Week*, the *Magazine of Wall Street*, *Moody's Investment Survey*, and other

[4] Wall Street Journal (1933).

publications were forecasting inflation. They cited Roosevelt's price-level objective. Several described a new "inflationary psychology."

Stock prices moved too. Jalil and Rua (2016) find that equities rose by an average of 5 percent in the 24 hours after unanticipated inflationary news. Industrial production, which had been falling since 1929, began to recover. From the trough in March 1933 through late 1933, output rose by about 15 percent. It then stalled around 10 percent above the trough as doubts about the regime grew.

As the stalling recovery signalled, Roosevelt had a problem: his administration was not united behind the inflation objective. Senior officials, including Director of the Budget Lewis Douglas and Acting Treasury Secretary Dean Acheson, resigned in protest against Roosevelt's price level objective (Davis 1986). More critically, the Federal Reserve was not initially on board. In September 1933, the Chicago Federal Reserve Bank's executive committee voted unanimously to reduce its participation in the open-market purchases designed to increase the monetary base and encouraged by the administration (Meltzer 2003). Chicago was supposed to take the largest share of the purchases, so its resistance mattered. Boston followed in November. Between November 1933 and March 1934, the Federal Reserve System stopped purchases altogether and Federal Reserve credit even declined slightly in 1934. Without Roosevelt's gold and silver purchases, the monetary base would have contracted (Meltzer 2003).

The result was a credibility problem. The Treasury and the Fed were not speaking with one voice. Inflation expectations began to falter in late 1933 (Jalil and Rua 2017). By the fall of 1933, one thing was clear: Roosevelt's price-level target required Federal Reserve cooperation.

### 2.1.2. Episode 2: Eccles Nominated as Federal Reserve Chair (November 1934)

Roosevelt's solution was institutional reform. He needed the Fed inside the regime. To achieve this, he appointed Marriner S. Eccles as Federal Reserve Chair. His nomination in November 1934 signaled that the Fed would no longer stand apart from the reflation program.

Eccles was an unusual choice. In February 1933, before Roosevelt took office and while serving as President of the Utah-based First Security Corporation, he testified before the Senate Committee on Finance, with a diagnosis that anticipated ideas Keynes would publish in the 1936 *General Theory*. Eccles blamed the Depression on "the vicious cycle of deflation": falling wages raised the real burden of debt, demand fell, and prices fell further still. The dollar, he said, was "too valuable measured in terms of goods and services.[5]"

[5] Eccles (February 13 to 28, 1933, p. 706).

Eccles wanted forceful government action. He proposed deficit-financed spending, infrastructure investment, farm debt relief, deposit insurance, and cancellation of wartime sovereign debts. Only then, he argued, could the government "bring about and support a price-level which will reestablish employment and credit." He also understood expectations. In his Senate testimony, he put the point bluntly:

> "[T]he entire public psychology must be changed through the propaganda of the press.[6]"

As Eccles understood, a regime change works only if people believe it.

Once appointed, Eccles moved quickly. He pushed for the Banking Act of 1935. The Act concentrated power at the Board of Governors and gave the President more authority over Federal Reserve governance (Kennedy 2003).[7] In testimony before the House Committee on Banking and Currency in February 1935, Eccles stated that:

> "The present need is to so modify our banking law as to encourage the banking system to give a full measure of cooperation to efforts at economic recovery.[8]"

The Banking Act did this. It centralized decision-making. It made it harder for regional Reserve Banks, such as Chicago and Boston, to block national policy. And the new power sat with Eccles, who supported accommodation.

This was a second regime shift. It did not announce a new target. It made the old target credible. Markets responded. From September 30, 1934, the month before Eccles's nomination, through July 31, 1935, the three-year Treasury yield fell by 126 basis points (Cecchetti 1988). The implication is simple. Markets expected short rates to stay at zero for much longer. Output then grew rapidly for nearly three years. By June 1937, industrial production had fully recovered to 3.0 percent above its pre-Depression peak.

### 2.1.3. Episode 3: The Mistake of 1937

After Eccles' appointment, the recovery was so strong that by 1937 policymakers became worried about inflation. Prices had not yet returned to the 1926 target. Still, the administration and the Federal Reserve began to fear that the recovery might go too far. The result was the "Mistake of 1937" (Eggertsson and Pugsley 2006).

The shift began in August 1936, when the Federal Reserve announced a 50 percent increase in reserve requirements, and gathered steam in December 1936, when the Treasury, led by Henry Morgenthau, started sterilizing gold inflows (Eggertsson and Pugsley

[6] *Ibid* (p. 720).

[7] While the Banking Act of 1935 is now central to Federal Reserve independence from Presidential influence (see Richardson and Wilcox (2025)), it concentrated power in the hands of Eccles who believed, as he argued during hearings for the Banking Act, that the Federal Reserve should be subservient to the executive branch.

[8] Eccles (1935, p. 184).

2006): instead of letting them swell bank reserves, it paid for the gold by selling debt, holding total reserves constant. On January 30, 1937, the Federal Reserve announced two further increases in reserve requirements, scheduled for March and May (Meltzer 2003).

The reserve-requirement changes were initially described as technical. Banks already held large excess reserves (Orphanides 2004).[9] But combined with gold sterilization, the signal was different. Policymakers were worried about inflation. They were willing to tighten. On February 18, 1937, Eccles stated publicly that short-term rates were excessively low and "there may be a tendency for rates near the vanishing point to increase." The *Wall Street Journal* reported the following day:

> "This is was the first time a member of the board has publicly described the reserve [requirement] increase as a device for preventing a further drop in long time rates.[10]"

This was forward guidance in reverse. The era of accommodation was ending and the era of active policy response was imminent. Rates would rise to prevent inflation. On March 15, Eccles told the *Chicago Daily Tribune*:

> "The upward spiral of wages and prices into inflationary levels can be as disastrous as the downward spiral of deflation.[11]"

According to the *Chicago Daily Tribune*, "Eccles' statement represent[ed] a complete reversal of the position that he and other financial advisers of the government [had] held[.]" Two weeks later, on April 2, Roosevelt confirmed in a press conference that concern about inflation was widespread within the administration:

> "I am concerned—we are all concerned—over the price rise in certain materials.[12]"

What earlier had seemed to be a "schism [...] within the ranks of the administration[,][13]" was now clearly a widely held view. The stock market fell 6 percent that day alone. Bond yields rose sharply. The three-year Treasury yield increased by 85 basis points from November 30, 1936, to March 31, 1937 (Cecchetti 1988). Expected time at the zero lower bound fell. Markets now expected accommodation to end sooner.

[9] Park and Van Horn (2015) and Calomiris, Mason, and Wheelock (2023) both find that the increase in reserve requirements did not materially affect bank lending because the large outstanding excess reserves meant that, in practice, few banks were bound by them. This is entirely consistent with our argument: it was not that the reserve requirements tightened monetary conditions when they were implemented, but rather that they signaled concern about inflation and a willingness to tighten conditions sooner than previously expected.

[10] Wall Street Journal (1937)

[11] Chicago Daily Tribune (1937a)

[12] Roosevelt (1937)

[13] Chicago Daily Tribune (1937b)

The macroeconomic consequences were severe. Output had stabilized near its pre-Depression peak. It then fell by 18 percent from July 1937 to May 1938, the sharpest peacetime contraction in recorded U.S. history (Eggertsson and Pugsley 2006). The Recession of 1937 was the mirror image of the recovery. It came from the unwinding of the reflation regime; a return to a more active policy stance.

### 2.1.4. Episode 4: The Reversal of 1938

The administration reversed course in 1938 once it was clear that the recovery had not been on as strong a footing as previously thought. Output had collapsed and the tightening had failed. On February 16, 1938, Roosevelt held a press conference with Treasury Secretary Morgenthau and Fed Chair Eccles at his side to reaffirm their commitment to accomodative policy(Eggertsson and Pugsley 2006). The *Chicago Daily Tribune* reported:

> "At his press conference today, the President said that he believes now, as he did in 1933, that achievement of permanent prosperity depends on raising general price levels to those prevailing in 1926.[14]"

Two days later, Roosevelt released a written statement prepared jointly by Morgenthau, Eccles, and economists from several executive departments:

> "It is clear that in the present situation a moderate rise in the general price-level is desirable[...]. Our program seeks a balanced system of prices such as will promote a balanced expansion of production.[15]."

Again, these were not only words. The Treasury ended gold sterilization on February 14, 1938, before the public announcement. The Federal Reserve reduced reserve requirements on April 18. The press understood the shift. An April 14 *New York Times* report stated:

> "The measures for expanding excess reserves which were announced on Thursday by President Roosevelt will recreate the bases for a great credit inflation.... Monetary management, after having been directed for some time towards guarding against a possible inflationary boom, has turned, under the pressure of the business depression, toward the other extreme.[16]"

The yield curve moved back. From January 31, 1938, to June 30, 1938, the three-year yield fell by 67 basis points (Cecchetti 1988). Output responded. From the May 1938 trough

[14] Chicago Daily Tribune (1938).
[15] Roosevelt (1938)
[16] New York Times (1938).

through December 1939, industrial production rose by 23 percent. This was before wartime spending dominated. Military spending did not accelerate significantly until late 1940 (Eggertsson 2008). The 1938–39 recovery was a return to the reflation regime, now with the Federal Reserve on board.

#### 2.1.5. Why These Four Episodes Represent a Predetermined Interest Rate Path

The effects of forward guidance depend crucially on the underlying policy regime. The regime during the Great Depression was passive: it looked approximately like a predetermined path for interest rates, which we show in section 3 should lead to large wings in output and inflation.

Three facts matter for this conclusion. First, the commitments were about price-level targets years into the future. They were not small near-term adjustments. Second, they were not conditioned on interim inflation or output realizations. There were no escape clauses. Third, short-term rates were near zero (Cecchetti 1988). The main margin was therefore the expected duration of zero rates. This is exactly the object in the forward-guidance experiment. For meaningful periods, especially March 1933 to late 1933 and late 1934 to early 1937, policy looked like a path without interim feedback.

This does not mean policy was literally a fixed peg at every date. It was not. Intermediate adjustments occurred. The Federal Reserve retained discretion. But markets appear to have believed the broad price-level objectives were binding. They also believed the central bank would accommodate those objectives with low rates for an extended period. That belief is enough for the mechanism. It is also enough for the classification: these episodes are close to *FG-Passive* in the sense relevant for the New Keynesian model.

### 2.2. Post-2008: Conditional Guidance with Escape Clauses

Post-2008 forward guidance was different. It was not an unconditional peg. Central banks said this explicitly. Their statements conditioned the rate path on inflation, slack, or other risks.

For instance, the Bank of Canada statement on April 21, 2009 made the path for interest rates explicitly conditional on inflation.

> "*Conditional on the outlook for inflation*, the target overnight rate can be expected to remain at its current level until the end of the second quarter of 2010[.][17]"

The Bank of Japan's statement on October 5, 2010 made the path for interest rates conditional on "risk factors" and financial imbalances.

[17] Bank of Canada (2009), emphasis added.

> "The Bank will maintain the virtually zero interest-rate policy until it judges [...] that price stability is in sight, *on condition that no problem will be identified in examining risk factors, including the accumulation of financial imbalances*.[18]"

The Federal Reserve statement on December 12, 2012 made the path for interest rates conditional on the unemployment rate, projected inflation, and longer-term inflation expectations.[19]

> "[T]his exceptionally low range for the federal funds rate will be appropriate at least as long as the *unemployment rate* remains above 6½%, *inflation between one and two years ahead* is projected to be no more than a half percentage point above the Committee's 2 percent longer-run goal, and *longer-term inflation expectations* remain well anchored.[20]"

The European Central Bank's statement on July 4, 2013, similarly, made the path for interest rates conditional on inflation.

> "The Governing Council expects the key ECB interest rates to remain at present or lower levels for an extended period of time. This expectation is *based on the overall subdued outlook for inflation* extending into the medium term[. . . .] In the period ahead, we will monitor all incoming information on economic and monetary developments and assess any impact on the outlook for price stability.[21]"

And the Bank of England's policy guidance on August 7, 2013 made the path for interest rates conditional on unemployment, projected inflation, medium-term inflation expectations, and financial stability.

> "[T]he MPC intends not to raise Bank Rate from its current level of 0.5% at least until the Labour Force Survey headline measure of the *unemployment rate* has fallen to a threshold of 7%[. . . .] The guidance [...] would cease to hold if [...] it is more likely than not, that *CPI inflation* 18 to 24 months ahead will be 0.5 percentage points or more above the 2% target; *medium-term inflation expectations* no longer remain sufficiently well anchored; [or . . .] the stance of monetary policy poses a significant threat to *financial stability*[.][22]"

[18] Bank of Japan (2010), emphasis added.

[19] The Federal Reserve's introduction of calendar-based forward guidance in August 2011 was similarly cautious, reading more like a prediction about the state of the economy rather than the announcement of a predetermined path for the interest rate or a shift in the policy rule: "The Committee currently anticipates that economic conditions–including low rates of resource utilization and a subdued outlook for inflation over the medium run–are likely to warrant exceptionally low levels for the federal funds rate at least through mid-2013." (Federal Open Market Committee 2011)

[20] Federal Open Market Committee (2012a), emphasis added.

[21] European Central Bank (2013), emphasis added.

[22] Bank of England (2013), emphasis added.

These were not pegs. Central banks wrote the escape clauses into the statements. If inflation threatened to overshoot, policy would respond. This is *FG-Active*. It is the regime under which the New Keynesian model predicts modest effects, consistent with the empirical literature (Campbell et al. 2019; Bundick and Smith 2020).

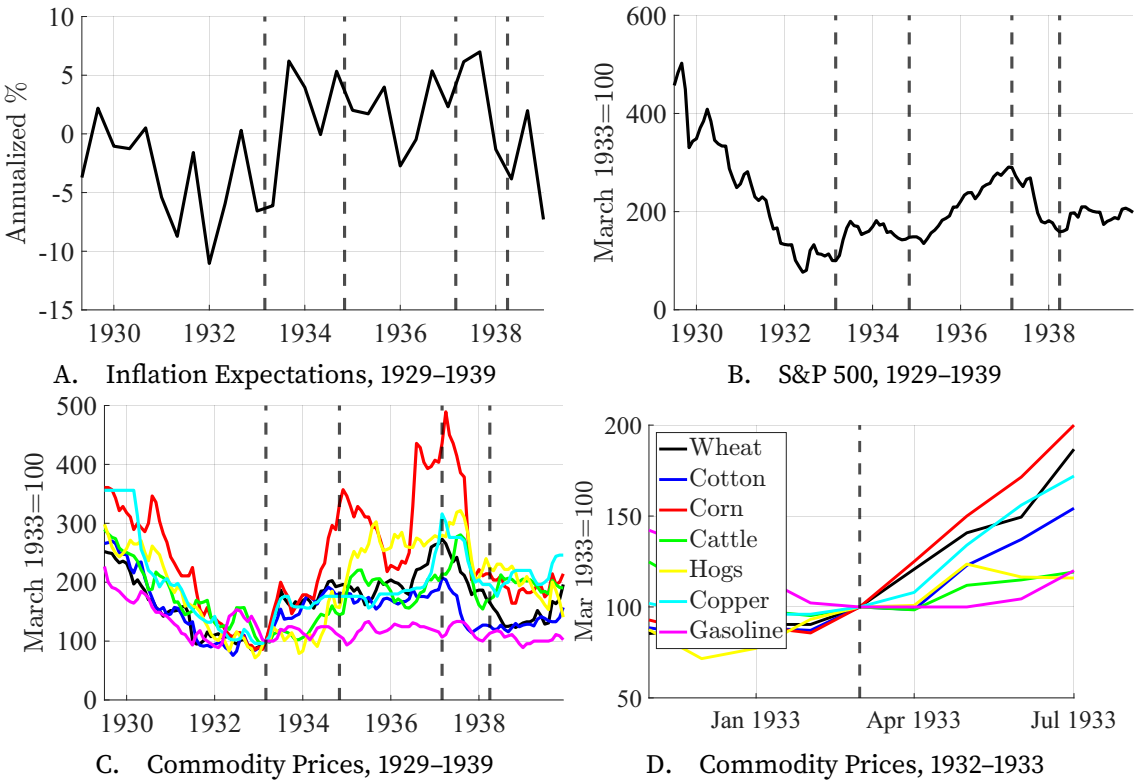
### 2.3. Expectations as a Mechanism: External Evidence from the Great Depressiong

The Great Depression announcements were about price-level regimes. They were not about inflation or output gaps conditional on future data. Roosevelt did not say, "we will restore 1926 prices unless inflation exceeds 2 percent in the interim." Eccles did not announce a Taylor-rule adjustment in 1937. He signaled that the accommodation regime was ending. The 1938 reversal was another regime change. It was a return to the price-level objective.

This is why the Depression is the better counterpart to the canonical forward guidance thought experiment. For meaningful periods, policy looked committed to a path that did not respond to interim inflation or output. Markets understood this. Treasury yields, stock prices, and commodity prices moved quickly. In Section 4, we show that the yield movements imply changes in expected zero-lower-bound duration of 14–37 months. Feeding those horizons into the model generates output and inflation movements of the same order of magnitude as in the data.

What is the mechanism leading to these large effects both in the data and in the model? It is expectations. Several sources show that Roosevelt's policies changed expected inflation. Hamilton (1992) uses commodity futures to infer inflation expectations. He finds that expected inflation had been negative in 1930–1932. It reversed sharply in spring 1933 and remained elevated through 1934–1936. The timing lines up with Roosevelt's inauguration and the departure from gold. The change in inflation expectations was large, on the order of 10 percentage points annualized, consistent with Roosevelt's announced price level objective (see Figure 2A). Cecchetti (1992) reaches a related conclusion using the term structure. The 1930–1932 deflation was largely anticipated and expected to continue. Roosevelt's 1933 announcement therefore generated a stark break from the old expected path. It was a new regime.

Jalil and Rua (2016) show the same thing in newspapers. Articles about inflation spiked in April 1933, after Roosevelt left gold. By late May, business analysts across *Business Week*, the *Economist*, *Moody's Investment Survey*, the *Magazine of Wall Street*, the *Review of Economics and Statistics*, and the *Trade and Money Index* were forecasting inflation. They cited Roosevelt's target. Several sources described rising "inflationary psychology." The change in inflation expectations rapidly boosted asset prices. Jalil and Rua (2016) find that equities rose by an average of 5 percent in the 24 hours after unanticipated inflationary news. Commodity prices also rose. Wheat, cotton, corn, cattle, hogs, copper, and gasoline all




A. Inflation Expectations, 1929–1939

B. S&P 500, 1929–1939

C. Commodity Prices, 1929–1939

D. Commodity Prices, 1932–1933

FIGURE 2. External Evidence on Inflation Expectations and Asset Prices

*Note*: External evidence on the mechanism. Top left panel: estimated expected inflation from commodity prices, Hamilton (1992). Top right panel: S&P 500 indexed to March 1933. Bottom left panel: commodity prices over the full period. Bottom right panel: commodity prices in after Roosevelt took office in March 1933.

jumped after Roosevelt took office (see Figure 2).

Eggertsson and Pugsley (2006) find similar evidence for 1937 and 1938 using the ProQuest Historical Newspaper Database. Articles about inflation rose from February to May 1937 and again from February to May 1938. The timing lines up with the policy announcements. Expectations changed in both directions: falling in 1937 and rising in 1938.

Pedemonte (2024) provides household evidence. He uses regional variation in radio exposure to study Roosevelt's April 1935 Fireside Chat. Areas with higher radio adoption increased bank debt and automobile spending and reduced savings in the two weeks after the speech, further evidence communication changed behavior.

Overall, a preponderance of evidence suggests a substantial change in inflation expectations during the early 1930s, driven by Roosevelt's policies and communications. The change in expectations, in the order of 10 percentage points annualized, reflects a significant shift in underlying monetary policy: a regime break from the gold standard era. By contrast, inflation expectations remained remarkably stable throughout the Great Reces-

sion.[23] By this point the Federal Reserve had established credibility in maintaining stable inflation and, if anything, sought to strengthen that credibility by announcing an explicit inflation target of 2 percent in 2012.

## 3. Analytics

Is the forward guidance puzzle a property of the New Keynesian model, or of a policy regime? This section shows it is the regime. The puzzle emerges under one assumption: that the central bank commits to a predetermined interest-rate path, suspending its usual response to inflation and output between the announcement of a future policy change and its implementation. We make this transparent through two experiments. The first shows that a future rate cut generates explosive responses when the interim path is predetermined (*FG-Passive*) but bounded responses when policy remains active (*FG-Active*). The second shows that the forward-guidance boom is the positive-sign mirror image of the Eggertsson and Woodford (2003) deflationary spiral—the same mechanism, opposite sign. The "puzzle" is a property of policy regimes, not of forward-looking behavior per se.

The policy regime during the Great Depression was *FG-Passive*, characterized by a commitment not to respond to, and even to encourage a large swing in inflation. The policy regime during the Great Recession, by contrast, was *FG-Active*, with central banks signaling they would actively respond to inflation and output while providing forward guidance about future policy.

### 3.1. The Model

We use the standard three-equation log-linear New Keynesian model to isolate the forward guidance mechanism.[24] The model consists of an IS equation, a Phillips curve, and a monetary policy rule:

$$\text{(IS)} \qquad y_t = \mathbb{E}_t y_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n)$$

$$\text{(PC)} \qquad \pi_t = \kappa y_t + \beta \mathbb{E}_t \pi_{t+1}$$

$$\text{(MPR)} \qquad i_t = \max\{0, \phi_\pi \pi_t + \phi_y y_t + r_t^n + \varepsilon_t\}$$

[23] Survey-based measures of inflation expectations from both households and professional forecasters indicate little variation in inflation expectations over this period. From January 2012 to January 2017, the Great Recession period when the Federal Reserve was most aggresive in its use of Forward Guidance, household one-year inflation expectations rose by at most 0.9 percentage points above average and professional forecasters' one-year expectations rose at most 0.2 percentage points above average. Household and forecasters' long-term inflation expectations rose by at most 0.2 percentage points over the same period. The response of inflation expectations was therefore several orders of magnitude smaller than during the Great Depression.

[24] See, for instance, Woodford (2003) and Galí (2015) for a standard textbook treatment.

Here $y_t$ is the output gap, $\pi_t$ is inflation, $i_t$ is the nominal interest rate, and $r_t^n$ is the natural real rate, and $\mathbb{E}_t$ is the expectations operator conditional on information available at time $t$. The monetary policy rule is subject to the zero lower bound (ZLB), imposed by the $\max\{\cdot, 0\}$ operator. The monetary policy shock $\varepsilon_t$ allows us to consider announcements of future rate cuts. In particular, we model future interest rate cuts as anticipated future shocks, $\mathbb{E}_t \varepsilon_{t+T}$.

Because both the (IS) and (PC) are forward-looking, the entire expected path of future real rates drives current demand and inflation. This makes forward guidance potentially powerful: promises about future policy affect today's decisions through expectations.

When the policy rate is strictly positive, the model has a unique bounded solution provided the Taylor principle holds:

$$\phi_\pi + \frac{1-\beta}{\kappa}\phi_y > 1 \tag{1}$$

This "active policy" condition ensures that the real rate rises in response to inflationary pressure, stabilizing the system. A violation of this condition corresponds to passive policy. The special case $\phi_\pi = \phi_y = 0$ represents a pure interest-rate peg.

For analytical tractability, we also work with a simplified version of the model that sets $\beta = 0$ in closed-form derivations, which allows us to obtain explicit solutions that clarify the economic mechanism. The Phillips curve then becomes

$$\pi_t = \kappa y_t, \tag{PC'}$$

and the IS equation collapses to:

$$y_t = (1 + \sigma\kappa)\mathbb{E}_t y_{t+1} - \sigma(i_t - r_t^n) \tag{IS'}$$

Numerical simulations use standard parameter values with $0 < \beta < 1$ to show that the basic mechanism carries through to more standard calibrations.

### 3.2. FG-Passive versus FG-Active: The Role of the Underlying Policy Regime

We revisit the canonical experiment to isolate the forward guidance channel in the simplest possible setting. Suppose the natural rate remains constant at its long-run steady-state value $r_t^n = r_L^n = -\log\beta > 0$ and the economy begins at the steady-state with $y_t = \pi_t = 0$. At time $t = 0$, the central bank announces a rate cut of size $\Delta$ to be implemented at time $T$:

$$\mathbb{E}_0 \varepsilon_T = -\Delta \tag{2}$$

The question is: how does the economy respond to this announcement at $t = 0$?

### 3.2.1. Cut Implemented Today

First, suppose the cut occurs immediately, at $T = 0$. In this case, the policy rate follows:

$$i_t = \begin{cases} \phi_\pi \pi_t + \phi_y y_t + r_t^n - \Delta, & t = 0 \\ \phi_\pi \pi_t + \phi_y y_t + r_t^n, & t > 0 \end{cases}, \tag{3}$$

and the impact is straightforward. With the policy rule active and the Taylor principle satisfied, the system returns to steady state after one period. The impact effect on output is:

$$y_0^{\text{now}} = \frac{\sigma \Delta}{1 + \sigma(\kappa \phi_\pi + \phi_y)} \tag{4}$$

This is the standard effect of a monetary policy shock: a rate cut lowers the real rate and stimulates demand through the familiar IS curve multiplier.

### 3.2.2. Future Cut with Predetermined Path

Now suppose the cut occurs at time $T > 0$, but—and this is critical—the central bank commits to holding the nominal rate fixed at some level $i^{\text{peg}}$ in all periods between $t = 0$ and $t = T$.[25] That is, policy follows:

$$i_t = \begin{cases} i^{\text{peg}}, & 0 \le t < T \\ \phi_\pi \pi_t + \phi_y y_t + r_t^n - \Delta, & t = T \\ \phi_\pi \pi_t + \phi_y y_t + r_t^n, & t > T \end{cases} \tag{5}$$

For simplicity, we set $i^{\text{peg}} = r_L^n$, so that the predetermined path coincides with the unique bounded solution absent any shocks to the policy rate: $\pi_t = y_t = 0 \Rightarrow i_t = r_L^n$.

Assuming the Taylor principle holds from time $t > T$ onward, $\mathbb{E}_T y_{T+1} = 0$. Our simplified $\beta = 0$ (IS') then implies that at time $T$:

$$y_T = y_0^{\text{now}} \tag{6}$$

Iterating backward, the impact effect at $t = 0$ is therefore:

$$y_0^{\text{FG-Passive}} = (1 + \sigma\kappa)^T y_0^{\text{now}} \tag{7}$$

[25] We show in Appendix B.1 that the intuition extends to any policy rule that is inactive in the sense that the Taylor principle does not hold: $\phi_\pi + (1/\kappa)\phi_y \le 1$.

Since $(1+\sigma\kappa) > 1$ and $y_0^{\text{now}} > 0$, output and inflation grow explosively with the horizon $T$. Pushing the cut outward magnifies the impact exponentially. This is the forward guidance puzzle.

The mechanism is intuitive once the predetermined path assumption is made explicit. Even though the interest rate peg coincides with the path for the policy rate absent any future shocks, it does not coincide with the normal policy rule. Instead, under a peg, the central bank commits not to respond to the evolution of output and inflation: this is what we term *FG-Passive*.

Think of a small spark in a building. If the firefighters respond immediately, the damage is limited. But if they commit not to intervene for $T$ periods—no matter how large the fire grows—the spark becomes an inferno. The forward-guidance boom operates identically: a small policy change, when combined with a credible commitment not to respond to the resulting inflation and output surge, snowballs into explosive dynamics. The longer the commitment not to act, the larger the boom.

### 3.2.3. The Role of the Peg

The explosive dynamics depend entirely on the predetermined path. If the central bank instead maintains its usual policy rule—responding to inflation and output according to its normal policy rule (MPR)—the large immediate response disappears.

Suppose that the central bank announces the same $\Delta$ cut to the policy rate at time $T$, but maintains its normal policy rule throughout, so that:

$$i_t = \begin{cases} \phi_\pi \pi_t + \phi_y y_t + r_t^n - \Delta, & t = T \\ \phi_\pi \pi_t + \phi_y y_t + r_t^n, & t \neq T \end{cases} \tag{8}$$

Substituting into (IS'), and iterating backward, the impact effect at $t = 0$ is:

$$y_0^{\text{FG-Active}} = \left( \frac{1+\kappa\sigma}{1+\sigma(\kappa\phi_\pi + \phi_y)} \right)^T y_0^{\text{now}} \tag{9}$$

Since $\kappa\phi_\pi + \phi_y > \kappa$ under the Taylor principle, the impact of a future shock now *diminishes* the farther into the future it occurs. The eigenvalue of the system lies inside the unit circle, and the system is stable. This illustrates a key point: it is not inherent to the New Keynesian model that a shock at $t+2$ has a larger effect than one at $t+1$. As long as the central bank follows an active policy rule, the opposite is true. What *is* inherent is that a two-period regime change moves the economy more than a one-period regime change.

Figure 3 confirms this result using conventional parameter values ($\beta = 0.99$, $\kappa = 0.1275$, $\sigma = 1$).[26] The solid blue lines show the explosive responses under *FG-Passive*, when the

[26] We use the calibration from Galí (2015), Chapter 3, as our baseline. The calibration is not crucial. Quali-

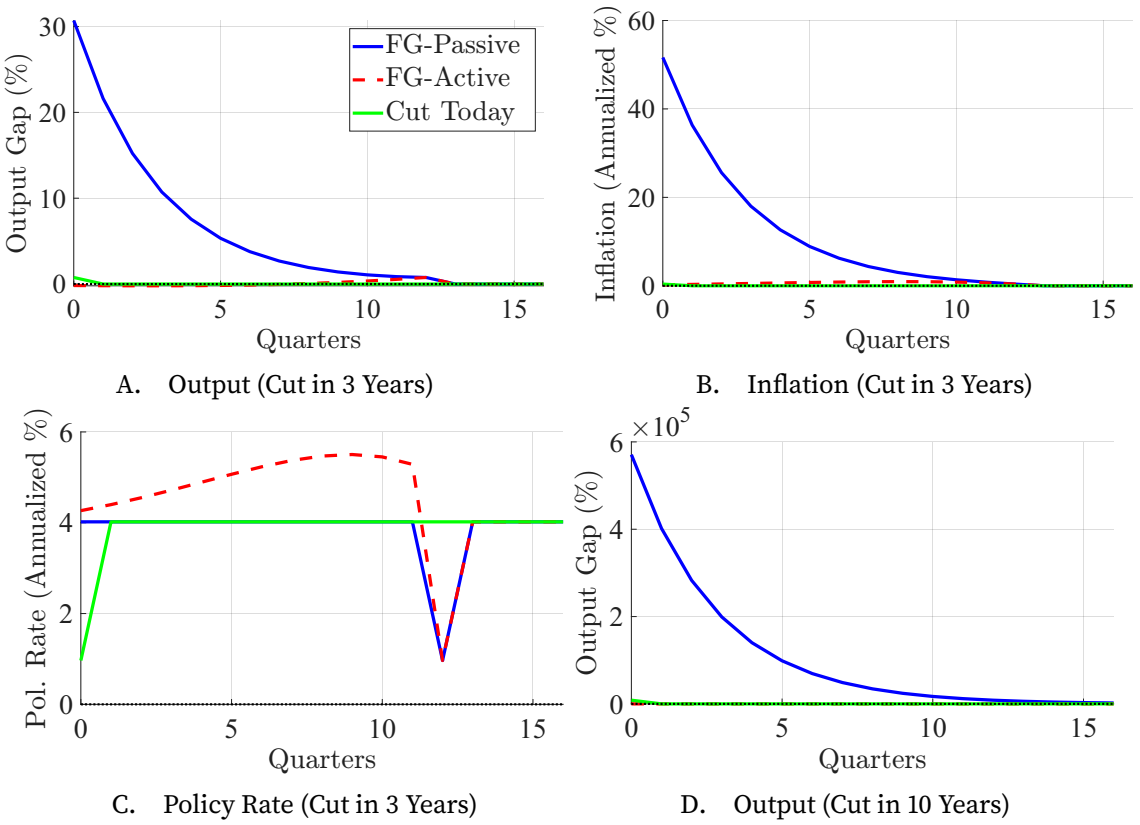



FIGURE 3. Responses to Forward Guidance: Predetermined Path vs Active Policy Rule

*Note*: Response of output, inflation, and the policy rate to a future policy rate cut. Blue line assumes a predetermined path for the policy rate ($\phi_\pi = \phi_y = 0$), while the dashed red line assumes an active policy rule satisfying the Taylor principle (we assume $\phi_\pi = 1.5$, $\phi_y = 0.5/4$, but the result does not depend on this assumption as long as the policy rule satisfies condition (1)). The green line presents the effects of the same policy rate cut today, at time $t = 0$.

interim path is predetermined. The dashed lines show the bounded responses under *FG-Active*, when the policy rule remains active throughout: a shock three years ahead has virtually no effect on impact. The difference between the two scenarios is dramatic. The predetermined path generates output and inflation booms that grow without limit as the horizon extends; the active rule produces responses that decay toward zero.

Figure 4 illustrates why the peg matters. It plots the cumulative "excess accommodation"—the difference between the pegged rate path and what the Taylor rule would prescribe. Inflation and output surge under the predetermined path, and the Taylor rule would call for aggressive rate increases to stabilize the economy. By committing not to respond, the central bank allows accommodation to compound over time. By contrast, under the active rule, the central bank raises rates immediately in response to the incipient boom, choking off the dynamics before they build. The puzzle is a straightforward implication

tatively similar results would hold for any reasonable parameter values.

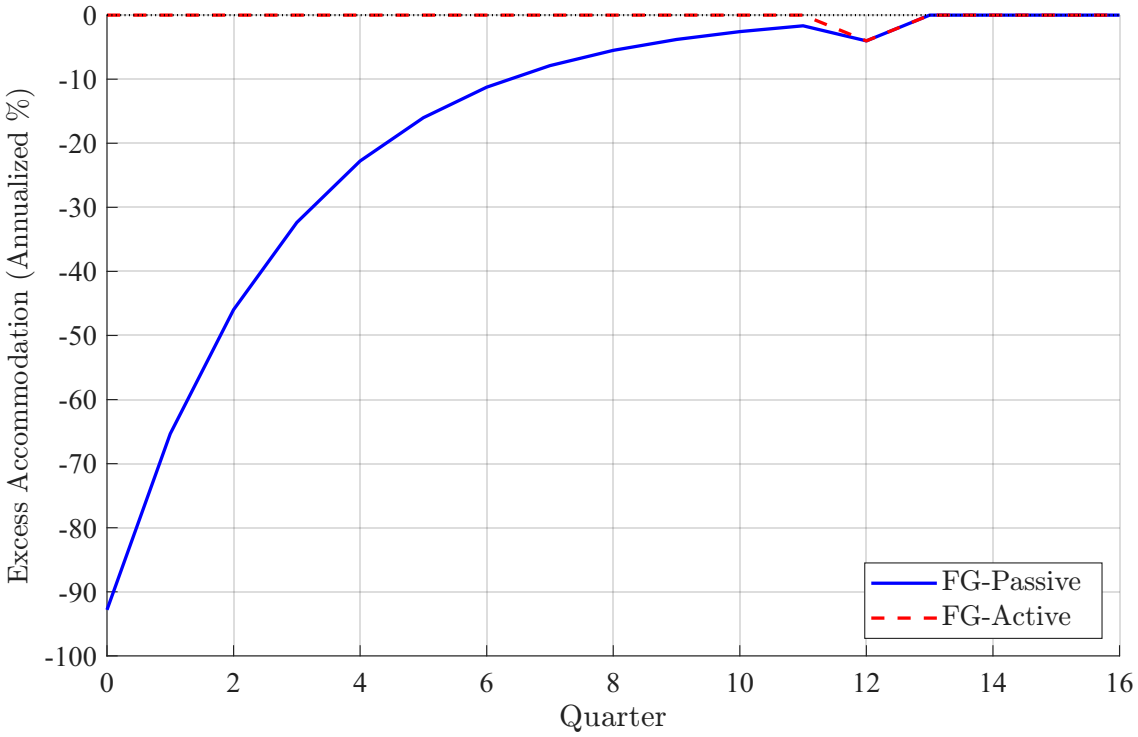



FIGURE 4. Excess Accommodation Under Predetermined Path vs Active Policy Rule

*Note*: The solid blue line shows excess accomodation from holding rates constat relative to a Taylor rule: how much lower the policy rate is in each period compared to where it would be if the central bank followed an active policy rule with the standard values of $\phi_\pi = 1.5$ and $\phi_y = 0.5/4$. The dashed red line shows the excess accomodation when the central bank follows an active policy rule in every period. In this case, the only deviation from normal policy is rate cut at time $T$.

of suspending the stabilization mechanism.

### 3.2.4. Comparison of Policy Regimes

To make the contrast between policy regimes concrete, Table 1 summarizes the key predictions. The *FG-Passive* generates effects that grow exponentially with the announcement horizon; the *FG-Active* generates effects that decay exponentially. We show in Appendix B.2 these results are not unique to future interest rate cuts: any anticipated shock to fundamentals (a "news shock") likewise produces explosive dynamics if policy is passive. By contrast, under an active policy rule news shocks generate negligible immediate effects because policy stabilization bounds expectations. The forward guidance puzzle is not specific to forward guidance; it results from any anticipated future shock as long as the policy regime suspends stabilization.

| Policy Regime | Announcement | Impact Effect (t=0) |
|---|---|---|
| Predetermined path | Future cut at $t = T$ | Explosive |
| Active rule (Taylor) | Future cut at $t = T$ | Negligible |
| Predetermined path | News shock at $t = T$ | Explosive |
| Active rule (Taylor) | News shock at $t = T$ | Negligible |

TABLE 1. Forward Guidance and News Shocks: Policy Regime Matters

### 3.3. Forward Guidance in a Liquidity Trap: The Deflationary and Inflationary Spiral

We now embed forward guidance into a liquidity trap environment, making the connection to Eggertsson and Woodford (2003) explicit. Suppose the natural rate drops to some $r_S^n < 0$ at time $t = 0$ and remains there for $\tau$ periods, after which it reverts permanently to $r_L^n > 0$.[27] The short-term nominal rate is constrained by the zero lower bound throughout, so $i_t = 0$ for all $t < \tau$. The question is: what happens to output and inflation while the natural rate is low?

#### 3.3.1. The Deflationary Spiral

With the zero lower bound binding and no forward guidance, the central bank cannot cut rates to offset the negative natural rate shock. The real rate rises, depressing demand. Because inflation and output are forward-looking, the collapse compounds over time through expectations. The IS equation for $t < \tau$ becomes:

$$y_t = (1 + \kappa\sigma)\mathbb{E}_t y_{t+1} + \sigma r_S^n \tag{10}$$

Iterating backward from $y_\tau = 0$ (as long as the Taylor principle is satisfied, the system jumps back to steady state when the natural rate recovers), we obtain:

$$y_t = \frac{(1 + \kappa\sigma)^{\tau - t} - 1}{\kappa} r_S^n \tag{11}$$

Because $(1 + \kappa\sigma) > 1$ and $r_S^n < 0$, output and inflation collapse explosively as the expected duration $\tau$ of the ZLB lengthens. Falling inflation raises the real rate, which further depresses demand, causing additional deflation in a self-reinforcing spiral. This is the Eggertsson–Woodford liquidity trap: the economy unravels because the central bank cannot offset the shock and expectations amplify the downturn.

[27] $r_S^n$ is the natural rate in the *short-run*, while the zero lower bound binds. $r_L^n$ is the natural rate in the *long-run*, after the zero lower bound no longer binds.

### 3.3.2. The Inflationary Spiral: Mirror Image

Now introduce forward guidance. Suppose that when the natural rate reverts to $r_L^n$ at time $\tau$, the central bank commits to holding the nominal rate at zero for an additional $T$ periods before returning to its active rule. The predetermined path now extends the ZLB into the recovery phase:

$$i_t = \begin{cases} 0, & 0 < t < \tau + T \\ \phi_\pi \pi_t + \phi_y y_t + r_L^n, & t \geq \tau + T \end{cases} \tag{12}$$

Once the natural rate has recovered (for $\tau < t \leq \tau + T$), the IS equation becomes:

$$y_t = (1 + \kappa\sigma)\mathbb{E}_t y_{t+1} + \sigma r_L^n \tag{13}$$

This is identical in structure to the deflationary case, except with $r_L^n > 0$ instead of $r_S^n < 0$. Iterating backward from $t = \tau + T$ with terminal condition $y_{\tau+T+1} = 0$ yields:

$$y_t = \frac{(1 + \kappa\sigma)^{\tau+T-t} - 1}{\kappa} r_L^n \tag{14}$$

Because $(1 + \kappa\sigma) > 1$ and $r_L^n > 0$, output and inflation now grow explosively with the forward guidance duration $T$. Rising inflation lowers the real rate, which stimulates demand, causing additional inflation in a self-reinforcing boom. The forward-guidance puzzle is the positive-sign mirror image of the deflationary spiral.

Figure 5 illustrates this symmetry visually. The left panel shows the deflationary spiral: output collapses as the expected ZLB duration $\tau$ lengthens. The right panel shows the inflationary spiral: output booms as the forward guidance duration $T$ extends. The two dynamics are governed by the same equation, differing only in the sign of the natural rate deviation from the predetermined nominal rate. Both arise when the short-rate path is predetermined—that is, when the central bank suspends interim feedback to realized inflation and output.

The parallel is not merely mathematical; it is economic. In the deflationary case, the central bank is *unable* to respond because the ZLB prevents rate cuts. In the inflationary case, the central bank *commits not to respond* despite being able to raise rates. The mechanism is otherwise identical: without interim stabilization, small shocks compound into large movements through forward-looking expectations. Eggertsson and Woodford (2003) analyzed the downward spiral; the forward-guidance literature rediscovered the upward spiral. The fundamental insight is the same: predetermined interest-rate paths generate explosive dynamics, whether through inability or commitment. There is however a key destinction: the forward-guidance boom can be eliminated at any moment by raising rates, whereas the deflationary spiral cannot be escaped, because the zero lower

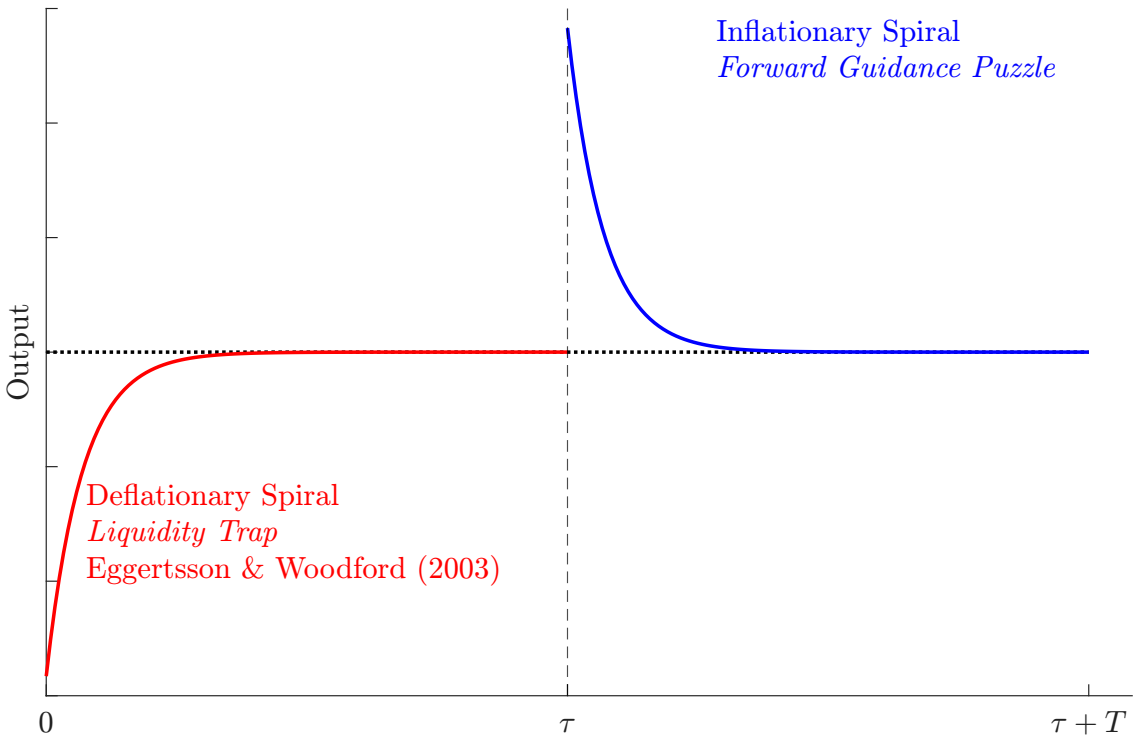



FIGURE 5. The Deflationary and Inflationary Spirals: Mirror Images

bound blocks the cuts that would offset it. Generating the forward guidance puzzle therefore requires a restrictive assumption—namely, that the central bank commits not to respond to the path of output and inflation despite having the ability to do so.

### 3.4. The Original Forward Guidance Puzzle: Combining the Spirals

The original forward guidance puzzle—the version that motivated the modern literature—combines the deflationary and inflationary spirals into a single episode. The economy is already at the ZLB due to a negative natural rate shock. The central bank then announces it will extend the ZLB duration beyond when the natural rate recovers, providing forward guidance about future monetary accommodation. How large are the effects on current output and inflation?

#### 3.4.1. Setup and Solution

We use the framework from subsection 3.3, but now solve for the full path. The natural rate is $r_S^n < 0$ for $\tau$ periods, then reverts to $r_L^n > 0$. Without forward guidance, the zero lower bound would bind until $t = \tau$ and policy would normalize afterward. With forward guidance, the central bank commits to holding $i_t = 0$ for an additional $T$ periods after the natural rate recovers, so the ZLB extends until $t = \tau + T$.

While the natural rate is low ($t \leq \tau$), output and inflation depend on expected output and inflation once it recovers. Without forward guidance, $\mathbb{E}_t y_\tau = \mathbb{E}_t \pi_\tau = 0$ (the economy jumps back to steady state when the natural rate recovers), and output collapses due to the deflationary spiral. With forward guidance, however, $\mathbb{E}_t y_\tau$ and $\mathbb{E}_t \pi_\tau$ are positive because the inflationary spiral continues for $T$ periods after $\tau$. These expectations propagate backward, offsetting—and potentially reversing—the deflationary drag. The solution for $t \leq \tau$ combines both spirals:

$$y_t = (1+\kappa\sigma)^{\tau-t+1} \frac{(1+\kappa\sigma)^T - 1}{\kappa} r_L^n + \frac{(1+\kappa\sigma)^{\tau-t} - 1}{\kappa} r_S^n \tag{15}$$

The first term is the forward-guidance boom, amplified backward through the expectations channel. The second term is the lingering deflationary drag from the negative natural rate. The multiplier $(1+\kappa\sigma)^{\tau-t+1}$ on the forward-guidance term shows how the boom at time $\tau$ is magnified exponentially as it propagates backward to time $t = 0$. The predetermined path during the ZLB acts as an amplifier. In subsubsection 3.4.2 we revisit the thought experiment in Del Negro, Giannoni, and Patterson (2023) that motivated the original forward guidance puzzle and show that the the explosive responses of output and inflation depend crucially on this multiplier: on the central bank committing not to respond to the path of output and inflation despite having the ability to do so.

Whether forward guidance generates unrealistically large effects in these circumstances is ultimately an empirical question, and hinges critically on the whether the Federal Reserve did—or is even able to–effectively communicate such a commitment. Campbell et al. (2019) find that the Federal Reserve's forward guidance during the Great Recession had limited and noisy effects on expectations; the New Keynesian model does not predict large responses of output and inflation once they account for these limitations. Bundick and Smith (2020) similarly find that Federal Reserve communication extended the expected duration of the ZLB by at most a few months. The models predictions when confronted with this limited forward guidance is in line with the data. We therefore turn to another period—the Great Depression—in section 4.

### 3.4.2. Quantitative Illustration and Decomposition

Figure 6 replicates the original forward guidance puzzle documented by Del Negro, Giannoni, and Patterson (2023). Their motivation is a striking figure from the New York Fed's DSGE model based forecasts in 2012. The model predicts that keeping the Federal funds rate at 25 basis points—their assumed effective lower bound—until 2015Q2, a 5-quarter extension of the ZLB beyond liftoff from the lower bound under baseline policy, leads to a jump in output growth to about 8½ percent. This jump in output occurs despite the fact that the ZLB extension implies only a modest deviation in the policy rate from its baseline.

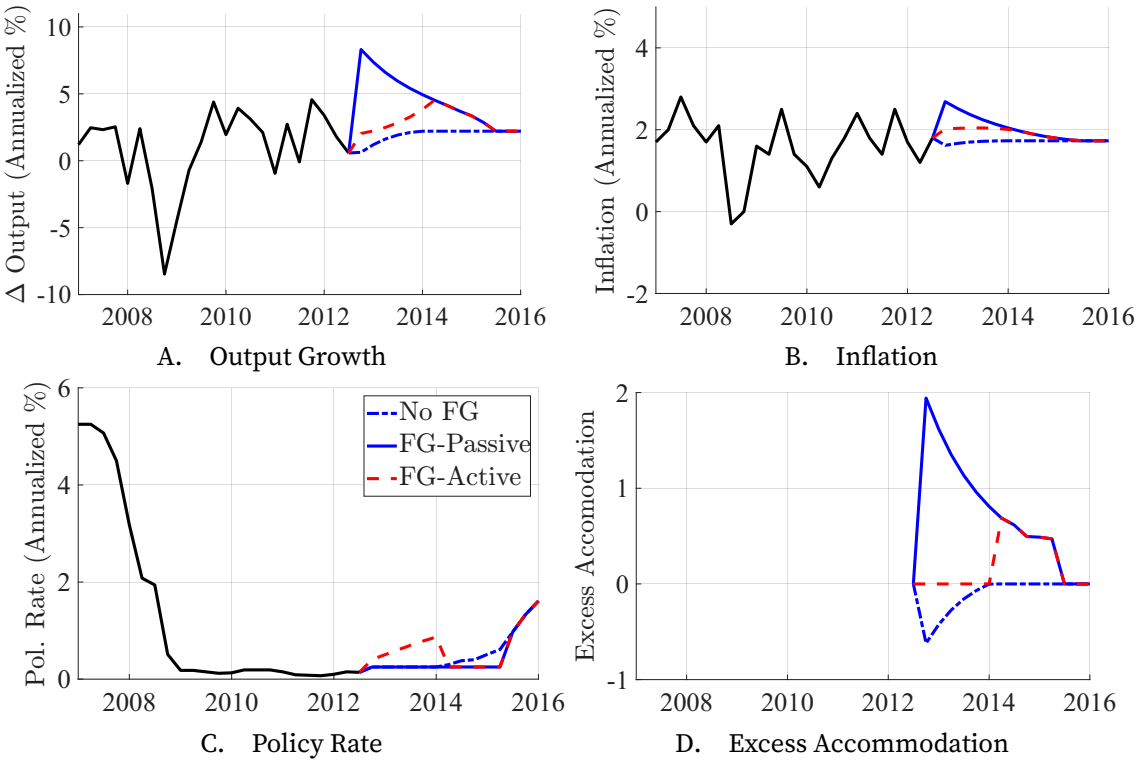



FIGURE 6. The Original Forward Guidance Puzzle: Replication of Del Negro et al. (2023)

*Note*: A replication of Del Negro, Giannoni, and Patterson (2023) Figure 1. The solid blue line is our calibration, designed to roughly replicate the paths of output growth, inflation, and the federal funds rate in the original figure when the central bank commits to keep the interest rate pegged at 25 basis points until 2015Q2. The dash-dotted blue line shows what our model predicts without forward guidance; the dashed red line shows the commitment to hold the rate at zero from mid-2014 to 2015Q2 but with active policy in all other periods.

Because the New York Fed's model predicts that the policy rate rises only gradually after 2014, the ZLB extension reduces the policy rate by just 20 basis points on average over its 5-quarter duration. The explosion in output in response to such a modest deviation in the policy rate from its expected path is, they argue, an unresonable prediction.

Our arguments above suggest that this prediction is perhaps not as unreasonable as it first appears. In the context of the motivational experiment in Del Negro, Giannoni, and Patterson (2023), a 5-quarter extension of the lower bound implies a central bank commitment to abandon its normal policy response function for 3-years—from mid-2012 to mid-2015—despite close to normal economic conditions, with inflation around 1.5 percent and real output growth around 2 percent in mid-2012. Much of the power of the forward guidance comes from this regime break in policy, rather than directly from the relatively minor deviation in the policy rate from its baseline path.

To demonstrate this, we calibrate the textbook New Keynesian model to roughly replicate the thought experiment in Del Negro, Giannoni, and Patterson (2023). Given the path

for the policy rate absent any forward guidance from 2014 onwards (the dashed blue line in Figure 6C), we choose $\beta$, $\kappa$, $\sigma$, and the path for the natural rate before 2014, to match the decline in output and inflation that the New York Fed's model predicts under baseline policy—absent any forward guidance—and the boom in output and inflation the model predicts with forward guidance, assuming that policy remains passive throughout. The dashed blue lines in Figure 6 plot our models predictions for output and inflation under baseline policy. The solid blue lines plot our models predictions assuming that the Federal Reserve commits to a 25 basis point policy rate peg through 2015Q2.

The dashed red lines in Figure 6 show what happens if we keep the 5-quarter extension but allow the Taylor rule to operate in all other periods—that is, we remove the predetermined path between announcement and liftoff. The boom all but disappears. By 2014Q1 the ZLB is no longer binding, and the model prescribes a policy rate about 86 basis points above zero. Figure 6D shows that removing the peg also removes most of the policy accomodation. The solid blue line plots excess accomodation assuming both forward guidance and a peg prior to 2014; the red dashed line plots the excess accomodation coming purely from the forward guidance. Removing the peg eliminates ¾ of the peak accomodation: the predetermined path between 2012 and 2014 drives the boom, not the 20-basis-point cut at the end.

We think of this alternative experiment as approximating a forward guidance announcement without a simulataneous commitment to change the underlying policy objectives. We believe the Federal Reserve's policy announcements in late-2012 are closer to this alternative thought experiment—forward guidance with the understanding that policy would remain responsive to the outlook for inflation and output—than to a pure policy rate peg.[28] The September 13, 2012 FOMC statement, for instance, announced that the Fed "anticipates that exceptionally low levels for the federal funds rate are likely to be warranted at least through mid-2015," but described this policy as "[c]onsistent with its statutory mandate [...] to foster maximum employment and price stability" and emphasized that "[t]he Committee will closely monitor incoming information on economic and financial developments in coming months."[29]

This decomposition clarifies the nature of the puzzle. The explosive effects arise because the central bank commits to tolerating massive overshooting in inflation and output for years without responding. It is not the announcement of future accommodation per se that generates the boom; it is the suspension of interim stabilization policy. The "puzzle" is that the model predicts—perhaps correctly—that such a radical regime change

[28] By this we don't mean that we think the Federal Reserve would literally have followed an interest rate path like the one in the red dashed line, but rather that communications around forward guidance always included exit clauses, and made policy explictly conditional on the economic outlook. This type of communication is inconsistent with the pure policy rate peg in the solid blue line as well. It's therefore perhaps not unsurprising that we didn't observe large responses in the data around the Great Recession.

[29] Federal Open Market Committee (2012b).

would have large effects. Once the regime change is made explicit, the result is no longer puzzling.

### 3.4.3. On Regime Reversals

A natural concern arises: if a regime shift to a predetermined path generates large positive effects, shouldn't the *reversal* of such a regime produce equally large negative effects? This question has been used to argue that the forward guidance puzzle implies implausibly volatile dynamics. The answer depends on expectations. If agents anticipate the reversal, its effects are already priced into earlier periods and produce no discrete surprise when the reversal occurs. In the experiments above, agents learn at $t = 0$ both that policy will be predetermined until $t = T$ *and* that the active rule will resume at $t = T$. The date-0 boom reflects both pieces of information; there is no sudden crash at $t = T$ because the reversion was anticipated throughout. Only *unexpected* regime reversals generate discrete negative shocks. As section 2 documented, the Great Depression featured several such surprises—the 1937 tightening came as the administration grew concerned about overheating; the 1938 resumption of accommodation reversed that shift—and the data show corresponding large swings in output. The mirror image logic holds, but only for unanticipated changes.

## 3.5. Summary: The Puzzle is Regime-Dependent

The analytics make the regime-dependence of the forward guidance puzzle transparent. When the central bank follows a predetermined interest-rate path—suspending interim feedback to inflation and output—the New Keynesian model generates explosive forward-guidance effects. These effects grow exponentially with the horizon, whether in a deflationary spiral, an inflationary boom, or the combination of the two in a liquidity trap with forward guidance. This is what we term *FG-Passive*, and the mechanism is the same in all cases: without stabilization policy, forward-looking expectations amplify small shocks into large movements.

When the central bank instead maintains a responsive policy regime—satisfying the Taylor principle and reacting to realized inflation and output under *FG-Active*—forward-guidance effects are bounded and generally modest. Future shocks matter less than current shocks because the policy response limits the buildup of inflationary or deflationary pressure. The system has a stable eigenvalue and does not explode.

This distinction matters because actual monetary policy regimes vary across time and context. As section 2 documented, the post-2008 forward guidance by modern central banks was explicitly state-contingent, with commitments conditioned on the inflation outlook and escape clauses built in. This is a responsive policy regime, and consistent

with the model's predictions for *FG-Active*, forward guidance effects during this period were moderate. By contrast, the U.S. Great Depression involved credible announcements of price-level targets that, for meaningful periods, appeared to commit policy to predetermined paths. The historical record shows large macroeconomic swings in response to these regime changes—exactly as the model predicts when the predetermined-path assumption is satisfied, as it is in *FG-Passive*.

The quantitative validation of this interpretation is the subject of section 4. We calibrate the model to the Depression-era decline and show that the model generates output and inflation responses on the same order of magnitude as those observed in the data in response to forward guidance durations of 14–37 months, in line with movements in the yield curve. The forward-guidance puzzle is therefore best understood as regime-dependent: large effects signal that the interim policy path was predetermined, while modest effects signal a responsive regime. The model works as intended when the policy regime is correctly classified.

## 4. Evidence from the Great Depression

In this section, we quantitatively examine the four events during the Great Depression discussed in detail in section 2. We conclude that the effects of forward guidance in the New Keynesian model are consistent with the data, at least in broad terms, once we examine a historical episode where *FG-Passive* is a good approximation of the policy stance. The model successfully reproduces the data for forward guidance durations that align with shifts in the anticipated ZLB duration, which we infer from changes in the 3-year yield using the expectations hypothesis of the term structure.

The pivotal events we study occur around the four highlighted dates in Figure 1:

1. Franklin D. Roosevelt's inauguration as President in March 1933, which ushered in the New Deal era, the elimination of the gold standard, and, crucially, a commitment to reflate prices to their 1926 level.
2. Marriner S. Eccles' accession to the Chair of the Federal Reserve Board in November 1934, which brought the Federal Reserve on board with the inflation objective, helping to break a crisis of confidence in the original price-level pledge.
3. The administration's premature retreat from Roosevelt's commitment to reflate prices in the mistake of 1937 leading to the sharpest recession in U.S. economic history.
4. The subsequent policy reversal in March 1938, when policymakers once again re-committed to reflation.

We adhere to the textbook New Keynesian model throughout—not because we believe it provides the best possible description of the Great Depression but because it offers a

TABLE 2. Forward Guidance by Episode: Yield Curve vs. Model

| Episode | 3-year yield change (bp) | Months of FG, $T_i$ | | Output (% from July 1929) | | |
|---|---|---|---|---|---|---|
| | | Yield curve | Matched | Yield curve | Matched | Data |
| 1. FDR takes office | −52 | 14 | 17 | −22.2 | −19.9 | −20.0 |
| 2. Eccles nominated | −44 | 36 | 36 | 3.0 | 3.0 | 2.9 |
| 3. Mistake of 1937 | 40 | 15 | 20 | −21.4 | −17.3 | −17.5 |
| 4. Reversal of 1938 | −45 | 37 | 37 | 4.7 | 4.7 | 5.4 |

*Note*: $T_i$ is the number of months the policy rate stays at zero after the natural rate returns to $r_L^n$, under the forward guidance (FG) in place after episode $i$; there is no FG in the trough, June 1932–March 1933. Yield-curve durations best match, under the expectations hypothesis, the change in the term-premium-adjusted 3-year Treasury yield (end of month) over March–May 1933, November 1934–February 1935, February–March 1937, and March–May 1938, moving $T$ from $T_{i-1}$ to $T_i$, with $T_0 = 0$. Matched durations bring model output closest to average output in the data over June 1933–November 1934, March–July 1937, May 1938, and November–December 1939. Output in the data is industrial production relative to July 1929, rescaled to average −30.0% over the trough.

benchmark for evaluating the New Keynesian model's predictions about forward guidance. The textbook model—the same model we use in section 3 but without the simplification that $\beta = 0$—features no discounting in the IS equation, no deviations from rational expectations, and no mechanism generating inertia in output or inflation. Output and inflation therefore respond especially strongly and quickly to announcements about future policy. In this sense, our results are conservative relative to alternatives that incorporate mechanisms that dampen the response of variables to forward guidance, including many of the recent extensions explored in the forward guidance puzzle literature.[30] Yet we show that even in this simple model forward guidance is not unrealistically powerful.

To preview our results, Table 2 demonstrates that the New Keynesian model predicts that regime changes under *FG-Passive* have large effects on output and inflation, and the data confirm that this is indeed the case. As we stress in the introduction, our central contention is not that the most basic version of the New Keynesian model provides the best fit to the data. Instead, it is that the strong reaction to forward guidance should not be considered puzzling in light of the data from the Great Depression which best corresponds to the forward guidance puzzle thought experiments in this literature.

We show in Appendix C that these results remain valid in a model with habit formation—allowing us to better match the more gradual changes in output observed in the data—and when we account for parameter uncertainty using Bayesian methods.

[30] Bundick and Smith (2020), for instance, show that a medium-scale New Keynesian model is consistent with the empirical responses of output and inflation to forward guidance during the Great Recession. We show that even a simple textbook New Keynesian model can generate responses of output and inflation to forward guidance that are broadly consistent with the data once we focus on a period of regime change.

### 4.1. Model

Our strategy is to write the simplest possible model capable of generating the drop in output from 1929 to 1933. We assume that the Great Depression resulted from a negative shock to the natural rate of interest of unknown duration.[31] The shock follows a two-state Markov process. The natural rate drops to $r_S^n < 0$ in 1929. Every month, there is a probability $\alpha$ that the natural rate reverts to $r_L^n = -\log\beta$, its long-run steady state value, and a probability $1-\alpha$ that it stays at $r_S^n$. Once the natural rate reverts to $r_L^n$, it stays there forever. We call the stochastic period when the natural rate reverts $\tau$.

In this setup, absent any forward guidance, output and inflation are the same in every period $t < \tau$ and jump to the steady state in period $\tau$. Adding forward guidance—a commitment to keep the nominal interest rate at zero for $T$ months after $\tau$—changes expected future output and inflation. Because of the forward-looking nature of the model, expected future output and inflation also effect output and inflation in $t < \tau$. Output and inflation in the low state are

$$y_t = \sigma\delta_y r_S^n + \alpha\delta_y \mathbb{E}_t y_\tau + \frac{\sigma\alpha}{1-\beta(1-\alpha)}\delta_y \mathbb{E}_t \pi_\tau, \tag{16}$$

$$\pi_t = \sigma\delta_\pi r_S^n + \alpha\delta_\pi \mathbb{E}_t y_\tau + \frac{\alpha\sigma\delta_\pi + \beta\alpha}{1-\beta(1-\alpha)}\mathbb{E}_t \pi_\tau, \tag{17}$$

where $\delta_y = \frac{(1-\beta(1-\alpha))}{\alpha(1-\beta(1-\alpha))-\sigma\kappa(1-\alpha)}$ and $\delta_\pi = \frac{\kappa}{1-\beta(1-\alpha)}\delta_y$.[32] Without forward guidance, $\mathbb{E}_t y_\tau = \mathbb{E}_t \pi_\tau = 0$. With forward guidance, we can solve backwards from the terminal condition $\mathbb{E}_t y_{\tau+T+1} = \mathbb{E}_t \pi_{\tau+T+1} = 0$ using Equations (IS) and (PC). Given $\alpha$, $\sigma$, $\kappa$, $r_L^n$, and $r_S^n$, we can use Equations (IS), (PC), (16), and (17) to calculate $y_t$ and $\pi_t$ for any given duration of forward guidance, $T$.

The two-state Markov process for $r_t^n$ implies that output and inflation in the low state are constant and adjust immediately to news about future output and inflation, creating a step-like pattern in response to changes in forward guidance, as in Eggertsson and Egiev (2025). Adding habit formation in consumption generates more gradual output responses to changes in forward guidance. We add a simple form of habit formation in the Appendix C.3 to show how the results are robust to using a model that can match the more gradual fall and recovery in output.

### 4.2. Calibration

To operationalize our Great Depression excersize we need to calibrate (1) the five model parameters ($\alpha$, $\beta$, $\kappa$, $\sigma$, and $r_S^n$) and (2) the forward guidance durations for each of the

[31] This is the only deviation from the model presented above in section 3, where we instead assume that the duration of the natural rate shock was known in advance.

[32] Note that, as in Eggertsson (2008), a bounded solution requires $\alpha(1-\beta(1-\alpha)) - \sigma\kappa(1-\alpha) > 0$.

TABLE 3. Calibration of the Great Depression Model

| Parameter | | Value | Calibration target |
|---|---|---|---|
| $\beta$ | Discount factor | 0.9986 | $r_L^n = -\log\beta$ |
| $\alpha$ | Probability that $r_t^n$ reverts to $r_L^n$ | 0.0513 | 3-year yield, end of February 1933 |
| $\kappa$ | Slope of the Phillips curve | 0.0015 | Inflation/output in the trough |
| $\sigma$ | Intertemporal elasticity of substitution | 1.1613 | Output in the trough |
| $r_S^n$ | Natural rate in the depressed state | −6.1663 | Natural rate, 1933 |
| $r_L^n$ | Long-run natural rate | 1.7132 | Natural rate, 1941–2000 |

*Note*: The model is monthly; natural rates are annualized percentages. $\beta$ is set so that $-\log\beta$ equals $r_L^n$, the average natural-rate estimate over 1941–2000, and $r_S^n$ is the 5th percentile of the natural-rate estimate for 1933 from Schmitt-Grohé and Uribe (2025). $\alpha$ matches the term-premium-adjusted 3-year Treasury yield on February 28, 1933 (0.92%) under the expectations hypothesis without forward guidance, which implies an expected 19.5 months at the zero lower bound. $\kappa$ and $\sigma$ are then set using inflation and output averages over the trough, June 1932–March 1933: −10.3% for year-over-year CPI inflation and −30.0% for output, measured by industrial production relative to July 1929 and rescaled to that decline. $\kappa$ is pinned down by the ratio of inflation to output; $\sigma$ is pinned downby the size of the downturn.

four regime changes ($T_1$, $T_2$, $T_3$, and $T_4$). We do this in two steps. First, we use external evidence on the natural rate and observed 3-year yields to infer $\alpha$, $\beta$, $r_n^s$, $T_1$, $T_2$, $T_3$, and $T_4$. Second, we use observed changes in output and inflation from 1929 to 1933 to infer $\kappa$ and $\sigma$. We report our calibrated parameters in Table 3, the calibrated forward guidance durations and model fit in Table 2, and show that our results are robust to parameter uncertainty in Appendix C.1 and Appendix C.2.

#### 4.2.1. Calibration using Treasury Yields

We start by calibrating $\alpha$, $\beta$, $r_S^n$, and the forward guidance durations $T_1$, $T_2$, $T_3$, and $T_4$ using data from the yield curve.

First, we use the natural rate estimates from Schmitt-Grohé and Uribe (2025) to pin down $r_L^n$, the long-run natural rate, and $r_S^n$, the short-run natural rate while the economy is in the Depression. We set $r_L^n = 1.71$ percent annualized, the 1941–2000 average of the estimates in Schmitt-Grohé and Uribe (2025). $r_L^n$ directly pins down $\beta$. In our monthly calibration, $\beta = \exp\left(-r_L^n/12\right)$.

We set $r_S^n = -6.16$ percent annualized, the 5th percentile of the 1933 estimates in Schmitt-Grohé and Uribe (2025). This choice for $r_S^n$ is consistent with the posterior median estimates in Denes and Eggertsson (2009) and with our belief that it should take a very large negative shock to the natural to generate a downturn as large as the Great Depression.

Given $r_L^n$, we then use the expectations hypothesis of the term structure to infer $\alpha$, $T_1$, $T_2$, $T_3$, and $T_4$ from 3-year yields, plotted in Figure 7. Because our approach implicitly assumes a constant term-premium, we adjust the raw 3-year yields using the regression

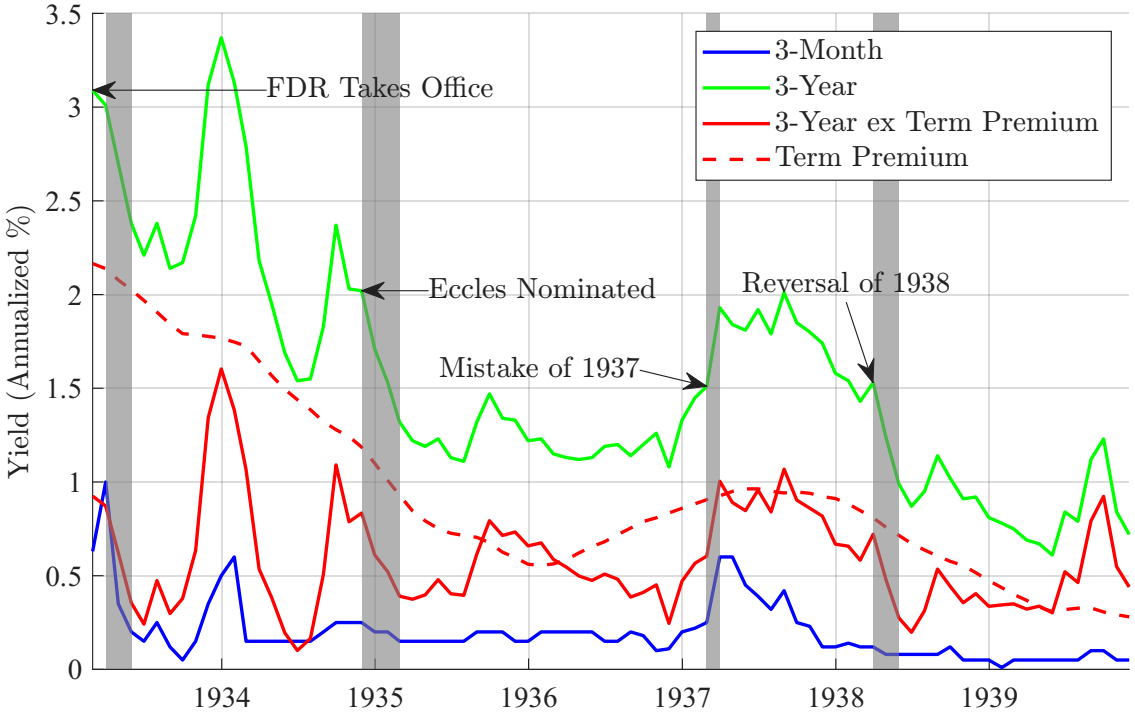



FIGURE 7. U.S. 3-Year Treasury Yield

*Note*: The solid green line reports raw 3-year yields from Cecchetti (1988). The dashed red line is the 24-month moving average of the term premium calculated from 3-month to 10-year yields using the regression based method suggested by Adrian, Crump, and Moench (2013). The solid red line is the term premium adjusted 3-year yield. The solid blue line reports the 3-month yield, a proxy for the short-term yield over this period. The shaded regions indicate periods we use to estimate changes in the expected duration of the ZLB. Source: Cecchetti (1988) and authors' calculations.

based approach introduced by Adrian, Crump, and Moench (2013) on the 3-month, 1-year, 2-year, 3-year, 4-year, 5-year, and 10-year yields for 1929 to 1949 reported in Cecchetti (1988). Since we only have monthly data and only for seven sparse yield durations, we subtract the smoothed 24-month moving average of our noisy term premium estimates from the raw 3-year yields to generate the solid-red 3-year yield ex term premium series in Figure 7.

We calibrate $\alpha$ using the term-premium-adjusted 3-year yield at the end of February 1933, prior to Roosevelt's inauguration on March 4, 1933. Since, as Jalil and Rua (2016) find, Roosevelt's inflationary policies were largely unexpected, we assume $T = 0$ at the end of February 1933 and calibrate $\alpha$ by solving

$$\min_{\alpha} \left( i_t^{3Y,\,\text{Data}} - i_t^{3Y,\,\text{Model}}(\alpha|\beta) \right)^2 \tag{18}$$

where $i_t^{3Y,\,Data}$ is the term-premium-adjusted 3-year yield at the end of February 1933 and

$$i_t^{3Y,\,Model}(\alpha|\beta) = \sum_{j=0}^{35}\left(1-(1-\alpha)^j\right)\frac{r_L^n}{36}. \tag{19}$$

Equation 19 expresses the term-premium-adjusted 3-year yield as the sum of the expected one month yields over the next 36 months, where the one month yield equals zero while $r_t^n = r_S^n$, and equals $r_L^n$ once the natural rate reverts to its long-run steady state. We find $\alpha = 0.0513$, implying that $r_t^n$ is expected to remain in its short-run depressed state for about 19.5 months.

Given $\alpha$, we can then compute the expected duration of forward guidance using changes in the the term-premium-adjusted 3-year yield in short windows around our four episodes. For each episode, we find the $T_i$ that solves

$$\min_{T_i}\left(\Delta i_t^{3Y,\,Data} - \Delta i_t^{3Y,\,Model}(T_i|\alpha,\beta)\right)^2 \tag{20}$$

where $\Delta i_t^{3Y,\,Data}$ is the change in the term-premium-adjusted 3-year yield and

$$\Delta i_t^{3Y,\,Model}(T_i|\alpha,\beta) = \sum_{j=0}^{35-T_i}\left(1-(1-\alpha)^j\right)\frac{r_L^n}{36} - \sum_{j=0}^{35-T_{i-1}}\left(1-(1-\alpha)^j\right)\frac{r_L^n}{36}. \tag{21}$$

Starting from the assumption that $T_0 = 0$ in February 1933, we can iteratively solve for $T_1$, $T_2$, $T_3$, and $T_4$.[33] The grey bars in Figure 7 indicate the periods we use to calibrate the expected forward guidance durations: March–May 1933, November 1934–February 1935, February–March 1937, and February–June 1939. Our assumption is that changes in the term-premium-adjusted 3-year yield during these windows primarily reflect changes in the expected path of the short-term interest rate due to changes in expected policy. We find $T_1 = 14$, $T_2 = 36$, $t_3 = 15$, and $T_4 = 37$.

Once we calibrate $\kappa$ and $\sigma$ as described below, we can also compute the expected durations of forward guidance needed for our model to match the observed path of output. For each episode we solve

$$\min_{T_i}\left(Y_t^{Data} - Y_t^{Model}(T_i|\alpha,\beta,\kappa,\sigma,r_S^n)\right)^2 \tag{22}$$

where we use Equation 16 to calculate $Y_t^{Model}(T_i|\alpha,\beta,\kappa,\sigma,r_S^n)$. We find that the model can match the observed path of output when $T_1 = 17$, $T_2 = 36$, $T_3 = 20$, and $T_4 = 37$. The solid black line in Figure 8A plots the model-implied output for these forward guidance

[33] If changes in yields imply more than 36 months of expected forward guidance, we extend the summation in Equation 21 by one month at a time until the expected forward guidance duration is fully captured.

durations. The dashed grey line plots the model-implied output for the forward guidance durations calibrated from 3-year yields.

#### 4.2.2. Calibration using Output and Inflation

We calibrate the model's remaining two parameters, $\kappa$ and $\sigma$, using observed output and inflation in the Great Depression.

Under our assumption that there was no forward guidance prior to Roosevelt's inauguration in March 1933, together Equation 16 and Equation 17 imply

$$\kappa = \frac{\pi_{1933}}{y_{1933}} \left(1 - \beta(1-\alpha)\right) \tag{23}$$

where $y_{1933}$ is output in the trough of the Great Depression (June–March 1933) relative to the pre-Depression peak, −30 percent, and $\pi_{1933}$ is the average inflation rate over the same period, −10.3 percent. This pins down kappa as 0.0015. This is close to the estimated slope of the Phillips curve in Hazell et al. (2022) of 0.0010 once converted from quarterly-unemployment to a monthly-output terms. The deflation of the 1930s and the post-war micro evidence—data that have nothing in common—point to the same flat Phillips curve. This agreement is a sign that the canonical model is not being forced.

We then calibrate $\sigma$ to match the size of the drop in output from 1929 to the trough in 1933. We solve

$$\min_{\sigma} \left( Y_t^{\text{Data}} - Y_t^{\text{Model}}(\sigma|\alpha, \beta, \kappa, r_S^n) \right)^2 \tag{24}$$

and find $\sigma = 1.1613$, once again not far from the standard values used in the literature.

In Appendix C.1 we show that our results do not depend on these exact parameters by jointly estimating the five parameters and the forward guidance durations using Bayesian methods.

### 4.3. Results

We now connect the basic results in Table 2 to the historical narrative to show that for each of the four events we consider, the models predictions are consistent with the responses of output and inflation in the data.

We conclude that once we examine a historical period where *FG-Passive* is a good approximation of the policy stance, the New Keynesian model does not overstate the power of forward guidance. The model predicts that a credible commitment to future inflation can have powerful effects on both output and inflation today. This is *borne* out in the data: both grew rapidly during the recovery from the Great Depression. In other words, regime changes have large effects in the model and in the data.

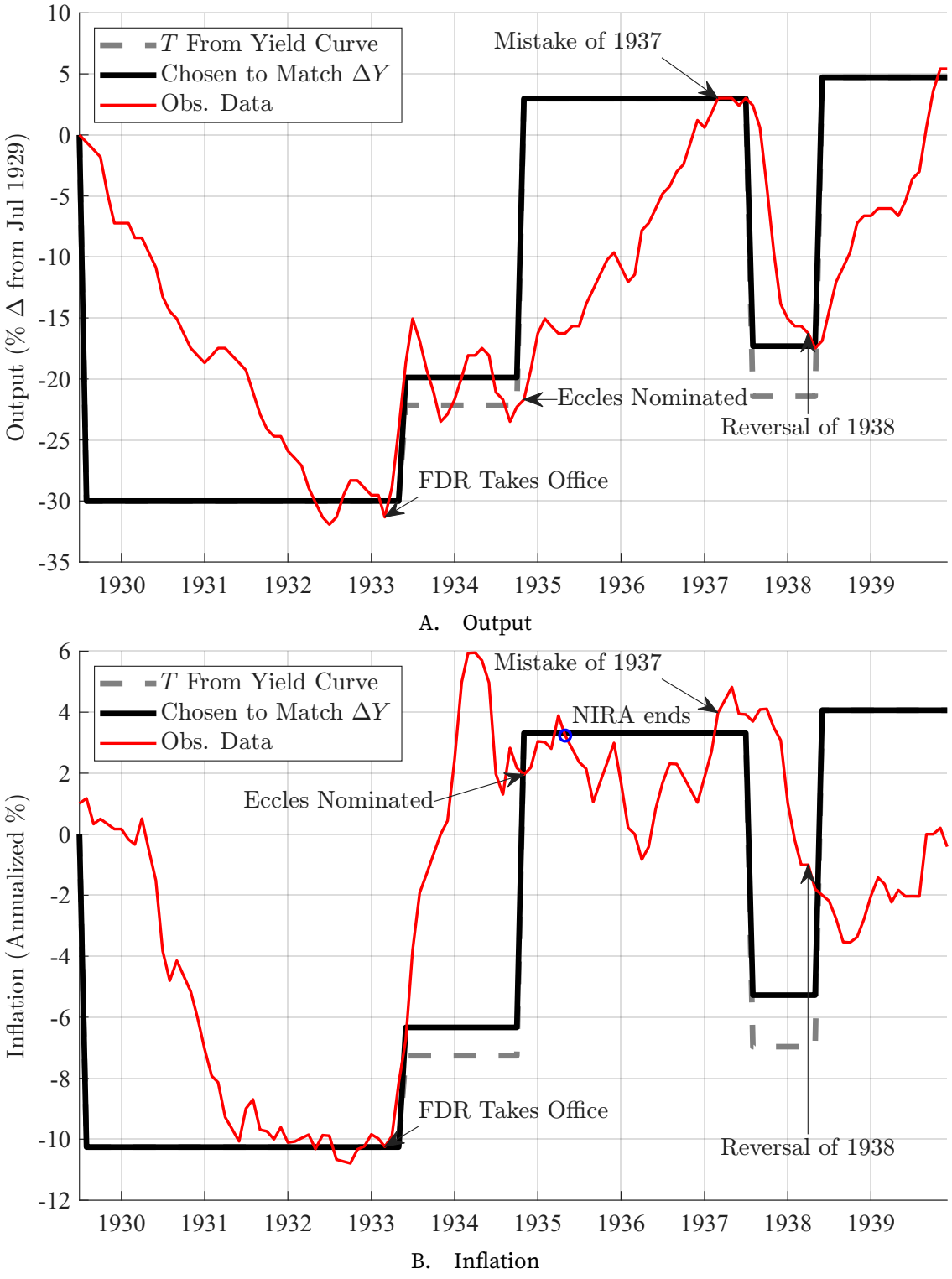



FIGURE 8. Response of Output and Inflation in Great Depression: Model Vs. Data

*Note*: Solid black lines are response of output (top) and inflation (bottom) to forward guidance picked to match response of output in the data to the regime changes. Dashed gray lines are are response of output (top) and inflation (bottom) to the amount of forward guidance suggested by the yield curve.

*Episode 1: Roosevelt takes office.* We first explore the response to Roosevelt taking office in March 1933. From March 31 to May 31, 1933, the term-premium-adjusted 3-year yield decreased by 52 basis points. According to our calibration, this decline implies about 14 months of forward guidance. In response, output initially jumped by 15 percentage points before stalling around 10 percentage points above the Great Depression trough.

In our calibration, 14 months of forward guidance increases output by 7.8 percentage points and inflation by 3.0 percentage points. Relative to the data, the model *understates* the response of output and inflation. We can match the 10 percentage point increase in output if the central bank commits to 17 additional months at the ZLB after the natural rate reverts to $r_L^n$. Setting $T_1 = 17$, the model predicts that output jumps by 10.1 percentage points and inflation rises by 4.0 percentage points, from -10.3 percent to -6.3 percent.

While the model's predictions for output are reasonable, forward guidance alone is insufficient to account for the large increase in prices from April 1933 to October 1934. This is hardly surprising in light of the existing literature. The Roosevelt administration introduced widespread policy changes that are difficult to capture with monetary policy alone. For instance, the National Industrial Recovery Act (NIRA), passed in June 1933, temporarily suspended antitrust laws to encourage firms to raise prices and facilitated the ability of unions to demand wage increases. Eggertsson (2012) shows that the NIRA can account for some of the rise in inflation during the initial recovery from the Great Depression. Inflation fell when the Supreme Court repealed the NIRA in May 1935—highlighted by the blue circle in Figure 8B.

In January of 1934, the Congress passed the Gold Reserve Act, allowing Roosevelt to raise the dollar price of gold by nearly 70 percent, and giving the Treasury broad authority to make purchases on both domestic and international markets to enforce the new price. This effective devaluation of the dollar may explain the additional rapid increase in prices in early 1934.

As we demonstrate in section 3, what matters for the power of forward guidance is whether households fully anticipated these jumps in inflation in March 1933, and whether, once they observed the increase in prices, they expected the inflation to persist. It is the expectation of future deviations from price stability that generate the large accumulation of monetary accomodation needed to generate an output boom. The relatively modest response of output in the data and the relative small amount of forward guidance implied by the yield curve suggest that households did not yet fully anticipate the inflationary regime shift. The response to Eccles' nomination suggests that Federal Reserve policy may have played a role in explaining why.

*Episode 2: Eccles' nomination.* By the fall of 1933, Roosevelt realized that the Federal Reserve's lack of cooperation and the dissent amoung its regional branches were undermin-

ing the credibility of the price-level objective. He responded by appointing a supportive Federal Reserve chair—Marriner S. Eccles—and enacting the Banking Act of 1935, which concentrated power at the Board, and therefore in Eccles' hands.

With the Federal Reserve now onboard, the inflationary regime shift became more credible. In response, from November 30, 1934, the month of Eccles's nomination, to February 28, 1935, the term premium adjusted 3-year yield decreased by a further 44 basis points—suggesting an additional 22 months of forward guidance, raising $T_2$ from 14 to 36 months. This shift in expectations ushered in nearly three years of sustained, rapid output growth. By mid-1937, output had fully recovered, surpassing its pre-Depression peak by 2.9 percentage points.

In our calibration, output rises to 3.0 percentage points above the pre-Depression peak and inflation rises to 3.3 percent following 36 month of forward guidance, a good match to the data on both output and inflation. In fact, 36 months of forward guidance is exactly what our model would pick to match the recovery in output following Eccles' appointment. Broadly speaking, we our model provides a good match to the output and inflation data eventhough we do not directly target inflation when choosing $T_2$.

*Episode 3: Mistake of 1937.* As the recovery from the Depression gained momentum, the Federal Reserve and the Roosevelt administration became increasingly concerned about potential runaway inflation, even though prices had not yet returned to their 1926 level. This led them to gradually withdraw their support for the previously announced price level targe, in what Eggertsson and Pugsley (2006) term the *Mistake of 1937*. When Roosevelt announced that "I am concerned—we are all concerned—over the price rise in certain materials[34]" in a press conference on April 2, 1937, the stock market fell by 6 percent.

Bond yields responded just as violently shortly after Eccles' announced that "there may be a tendency for rates near the vanishing point to increase[35]" on February 18, 1937. The term-premium-adjusted 3-year yield rose by 40 basis points from February 28, 1937, to March 31, 1937, suggesting about a 21 month reduction in forward guidance—a drop in $T_3$ from 36 to 15 months. Output, which had stabilized near the pre-Depression peak, plummeted by 17.5 percentage points from July 1937 to May 1938, ushering in the second phase of the Great Depression.

In our calibration, output falls by 21.4 percentage points and inflation falls by 10.2 percentage points when forward guidance declines from 36 months to 15 months. The model can replicate the output response when we instead reduce the central bank's forward guidance commitment by 16 months, decreasing $T_3$ from 36 to 20. With $T_3 = 20$, the model predicts a 17.3 percentage point output decline and a drop in inflation from 3.3

[34] Roosevelt (1937)

[35] EcclesSaysReserve1937

percent to -5.3 percent.

*Episode 4: The reversal of 1938.* The Roosevelt administration reversed course following the rapid decline in output and inflation in late 1937 and early 1938, re-committing to the inflation objective.

Just as the retraction of the inflation objective led to a rapid increase in Treasury yields, the re-establishment of the commitment in a press conference on February 16, 1938 led to a substantial decline in 3-year yields. From March 31, 1938, to May 30, 1938, the term-premium-adjusted 3-year yield decreased by 45 basis points, suggesting that forward guidance rose by about 22 months—from 15 to 37 months. In response, from May 1938 to December 1939, output rose by 22.9 percentage points, to 5.4 percentage points above the pre-Depression peak. The turning point in the Recession of 1937 aligns more closely with this renewed regime change than with the later expansion of World War II spending. A common misconception is that World War II ended the Great Depression. This glosses over the incredible recovery in output between 1933 and 1937—the period with the fastest growth in U.S. peacetime history—and the recovery from the second phase of the Depression, which took hold before the start of the World War, thanks to a return to the economic policies of the 1933 to 1937 period.

In our calibration, output rises to 4.7 percentage points above the pre-Depression high and inflation rises from -6.9 percent to 4.1 percent when forward guidance rises from 15 months to 37 months, a good fit to the response of output. Once again, 37 months of forward guidance is exactly what our model would pick to match the recovery in output following the reversal of 1938.

*The Effectiveness of Forward Guidance Revisited.* Overall, the results across all four episodes suggest that the Great Depression was indeed a period where *FG-Passive* provides a good approximation to the policy stance, and once we focus on such a period, the New Keynesian model no longer over-predicts the effectiveness of forward guidance. Instead, the model's predictions are of the same order of magnitude as the responses of output and inflation observed in the data. We demonstrate that these findings are robust to alternative parameter values in Appendix C.2.

## 5. Conclusion

We argue, both theoretically and empirically, the forward guidance puzzle is regime-dependent. Theoretically, the puzzle arises when monetary policy commits to a major regime change. In the typical thought experiment, the central bank commits not only to a future interest rate shock but also to staying on that path regardless of how inflation and

output evolve between now and the time of the shock. Generating explosive paths for output and inflation requires an explosive amount of additional accommodation, and there is little reason to expect a central bank not to react in normal circumstances. Indeed, when central banks made use of forward guidance during the Great Recession, their communication signaled a continued commitment to their normal inflation objectives, not an unconditional policy peg. We argue that this implies that major policy regime changes, and not the nuanced forward guidance we saw during the Great Recession, are the appropriate empirical counterpart to the forward guidance puzzle.

There were, however, four major regime breaks in the Great Depression, that we believe represent an appropriate empirical counterpart to the theoretical puzzle. For each of these four regime changes, the large effects of forward guidance predicted by the New Keynesian model are of the same order of magnitude as those observed in the data. The New Keynesian model predicts that regime changes have large effects on output and inflation. The data confirm that output and inflation rose (or fell) dramatically in response to actual regime changes during the Great Depression, mirroring the model's predictions. In this sense, the forward guidance puzzle is a feature of the model that allows it to match large changes in output and inflation in response to economic regime changes, rather than a bug. Our findings suggest that that forward guidance can have sizable impacts on output and inflation if it signals a true and credible regime change in monetary policy.

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## Appendix A. Comparison of Industrial Production with GDP and GNP

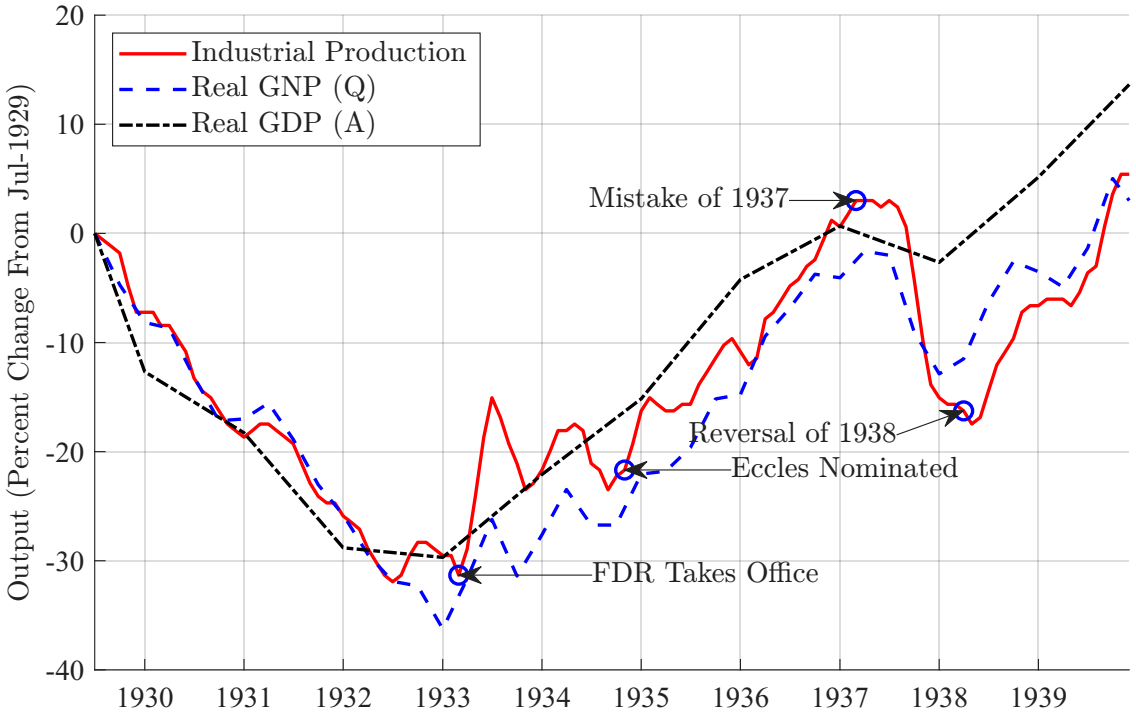



FIGURE A1. Industrial Production, GDP, and GNP During the Great Depression, 1929–1939.

*Source*: Industrial production from FRED (Industrial Production: Total Index (INDPRO)); GDP from FRED (Real Gross Domestic Product (GDPCA)); GNP from Balke and Gordon (1986).

## Appendix B. Additional Results on Forward Guidance Thought Experiment

### B.1. Extending Intuition to General Inactive Policy Rules

In section 3, we argue that the forward guidance puzzle is an artifact of the policy regime rather than the nature of forward-looking models themselves. Our analytical results compare the effects of announcing a future interest rate cut under either a predetermined path for the policy rate or an active policy rule (such as a Taylor rule). We show that under a predetermined path, the announcement generates explosive dynamics, whereas under an active rule, the immediate impact is negligible.

Here we will show that the argument extends beyond predetermined paths to any general inactive policy rule. As long as policy does not respond strongly enough to the path of inflation and output, the model predicts that future interest rate cuts generate explosive dynamics. In this sense, the destinction really is between *FG-passive*, a regime

where policy violates the Taylor principle, and *FG-active* a regime where policy adheres to the Taylor principle, and therefore stabilizes expectations.

Suppose that the central bank announces a policy rate cut at date $T$, $\varepsilon_T = -\Delta$, and that monetary policy follows

$$i_t = \begin{cases} \phi_\pi \pi_t + \phi_y y_t + r_t^n & 0 < t < T \\ \phi_\pi^{\text{Post}} \pi_t + \phi_y^{\text{Post}} y_t + r_t^n - \Delta & t = T \\ \phi_\pi^{\text{Post}} \pi_t + \phi_y^{\text{Post}} y_t + r_t^n & t > T \end{cases} \tag{A1}$$

Where $\phi_\pi^{\text{Post}} + \frac{1}{\kappa}\phi_y^{\text{Post}} > 1$—the post-announcement policy rule satisfies the Taylor principle—and $0 \leq \phi_\pi, \phi_y$ are positive but do not necessarily satisfy the Taylor principle. As long as $\phi_\pi^{\text{Post}}, \phi_y^{\text{Post}}$ ensure determinacy from time $T$ onwards, we can solve backwards from the terminal condition $y_{T+1} = \pi_{T+1} = 0$.

At time $T$, the effect of the rate cut is

$$y_T = \frac{\sigma\Delta}{1 + \sigma(\phi_y^{\text{Post}} + \kappa\phi_\pi^{\text{Post}})} = y_0^{\text{Now}}. \tag{A2}$$

Solving backwards,

$$y_0 = \left[\frac{1 + \sigma\kappa}{1 + \sigma(\phi_y + \kappa\phi_\pi)}\right]^T y_0^{\text{Now}} \tag{A3}$$

Whenver the Taylor principle is satisfied, $\kappa\phi_\pi + \phi_y > \kappa$,

$$\left[\frac{1 + \sigma\kappa}{1 + \sigma(\phi_y + \kappa\phi_\pi)}\right] < 1.$$

The impact of forward guidance diminishes the further in the future it occurs. Conversely, whenever the Taylor principle is violated, $\kappa\phi_\pi + \phi_y < \kappa$,

$$\left[\frac{1 + \sigma\kappa}{1 + \sigma(\phi_y + \kappa\phi_\pi)}\right] > 1.$$

The impact of forward guidance increases the further in the future it occurs. The forward guidance puzzle appears whenever policy is passive—whenever the central bank does not respond aggresively enough to inflation and output fluctuations to stabilize expectations.

Figure A2 confirms that the same intuition holds in the full model with $0 < \beta < 1$. The black line in Figure A2 represents the boundary where the Taylor Principle holds. At points below the line, the Taylor Principle is violated and policy is passive. At points above the line, the Taylor Principle holds and policy is active. In the red region, forward guidance shocks have a larger effect on output than the same shock today—the economy

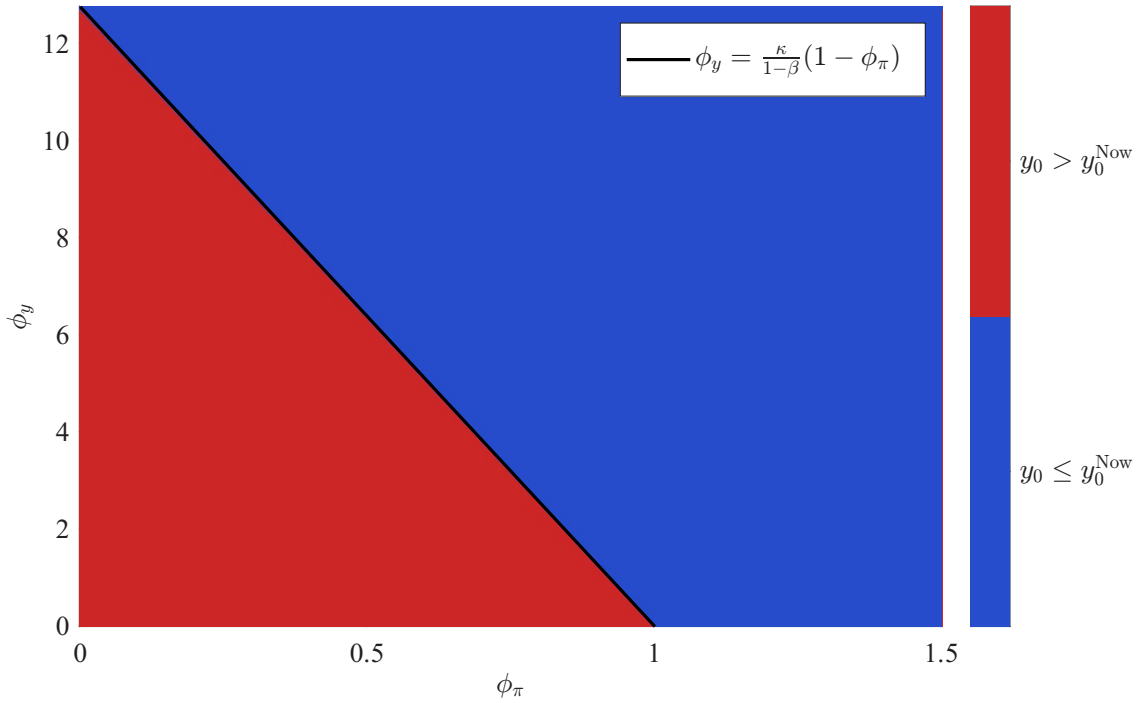



FIGURE A2. Response of Output to Forward Guidance Shock vs. Same Shock Today

*Note*: Plots the response of output on impact, at $t = 0$, to a forward guidance shock at time $T$ compared to the response of output to the same size shock to the policy rate today for various policy rules (different values for $\phi_\pi$ and $\phi_y$). Other parameter values: ($\beta = 0.99$, $\sigma = 1$, $\kappa = 0.1275$). The black line plots the boundry of the region where the Taylor Principle holds. At points below the black line, the Taylor Principle is violated and policy is passive. At points above the black line, the Taylor Principle holds and policy is active. In the red region, output responds more to an anticipated future shock than to a shock today: the explosive dynamics dominate. In the blue region, output responds less to an anticipated future shock than to a shock today: policy stabilization dominates.

features explosive dynamics. In the blue region, forward guidance shocks have a smaller effect on output than the same shock today—policy is active enough to stabilize expectations. The red region overlaps perfectly with the region where the Taylor principle is violated, whereas the blue region is precisely the determinacy region where the Taylor principle holds.

### B.2. Results for News Shocks

The forward guidance puzzle phenomenon is not unique to news about future monetary policy. In this section we show that similar dynamics arise in response to news shocks about future economic fundamentals. We focus specifically on news about future cost push and demand shocks.

To start, we add two additional shocks to the standard three-equation New Keynesian

model: a cost push shock $\mu_t$ and a demand shock $z_t$.

$$y_t = \mathbb{E}_t y_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} + r_t^n) + z_t, \tag{A4}$$

$$\pi_t = \beta \mathbb{E}_t \pi_{t+1} + \kappa y_t + \mu_t, \tag{A5}$$

$$i_t = \max\left\{0, \phi_\pi \pi_t + \phi_y y_t + r_t^n + \varepsilon_t\right\} \tag{A6}$$

We want to think about the effects today of news shocks about future cost push and demand shocks under different monetary policy rules. For analytical tractability and consistency with the results in section 3, we start with the $\beta = 0$ case. In this case we can write,

$$\pi_t = \kappa y_t + \mu_t \tag{A7}$$

$$y_t = (1 + \sigma\kappa)\mathbb{E}_t y_{t+1} - \sigma(i_t - r_t^n) + \sigma \mathbb{E}_t \mu_{t+1} + z_t \tag{A8}$$

We will also assume that the policy rule ensures the Taylor Principle holds from time $T$ onwards. Monetary policy follows

$$i_t = \begin{cases} \phi_\pi \pi_t + \phi_y y_t + r_t^n & 0 < t < T \\ \phi_\pi^{\text{Post}} \pi_t + \phi_y^{\text{Post}} y_t + r_t^n & t \geq T \end{cases} \tag{A9}$$

Where $\phi_\pi^{\text{Post}} + \frac{1}{\kappa}\phi_y^{\text{Post}} > 1$—the post-announcement policy rule satisfies the Taylor principle—and $0 \leq \phi_\pi, \phi_y$ are positive but do not necessarily satisfy the Taylor principle.

### B.2.1. Demand Shock

First, we consider the effect of an anticipated demand shock at time $T$, $z_T = \Delta$. Solving backward from the terminal condition $y_{T+1} = \pi_{T+1} = 0$, the response of output at time $T$ is

$$y_T = \frac{\Delta}{1 + \sigma(\phi_y^{\text{Post}} + \kappa\phi_\pi^{\text{Post}})} = \frac{1}{\sigma} y_{Post}^{Now} = y_0^{\text{Now, D}} \tag{A10}$$

Solving backwards,

$$y_0^{Demand} = \left[\frac{1 + \sigma\kappa}{1 + \sigma(\phi_y + \kappa\phi_\pi)}\right]^T y_0^{\text{Now, D}} \tag{A11}$$

Whenever the Taylor principle does not hold, $\phi_y + \kappa\phi_\pi < \kappa$, news about future demand shocks has larger effects on output today than on impact. The intuition is identical to the forward guidance case; indeed the expression for the response of output is identical up to a multiplicative constant. Whenever the Taylor principle does hold, policy stabilizes

expectations, and news about future demand shocks no longer has explosive effects in the present.

### B.2.2. Cost Push Shock

Now consider the effect of an anticipated cost push shock at time $T$, $\mu_T = \Delta$. Solving backward from the terminal condition $y_{T+1} = \pi_{T+1} = 0$, the response of output at time $T$ is

$$y_T = -\frac{\sigma\phi_\pi^{\text{Post}}\Delta}{1+\sigma(\phi_y^{\text{Post}} + \kappa\phi_\pi^{\text{Post}})} \tag{A12}$$

The response of output at $T-1$ is

$$y_{T-1} = \frac{1+\sigma\kappa}{1+\sigma(\phi_y+\kappa\phi_\pi)} y_T + \frac{\sigma\Delta}{1+\sigma(\phi_y+\kappa\phi_\pi)} \tag{A13}$$

Solving backward, the response of output at time 0 is

$$y_0^{CostPush} = \left[\frac{1+\sigma\kappa}{1+\sigma(\phi_y+\kappa\phi_\pi)}\right]^{T-1} y_{T-1} \tag{A14}$$

Once again, whenever the Taylor principle does not hold, $\phi_y+\kappa\phi_\pi < \kappa$, news about future cost push shocks has generates explosive dynamics.

### B.2.3. Confirmation of Results in Full Model

We confirm that the qualitative results obtained in the simplified model carry over to the full model. Specifically, the effects of anticipated demand and cost push shocks on current output are amplified when the Taylor principle does not hold, and are stabilized when the Taylor principle is satisfied. Figure A3 plots the responses of output and inflation to anticipated future demand and cost push shocks assuming that $\beta = 0.99$, $\sigma = 1$, $\kappa = 0.1275$.

## Appendix C. Robustness of Great Depression Results

### C.1. Bayesian Estimation of Parameters and Forward Guidance

The calibration in subsection 4.2 pins down each parameter with a single piece of evidence and then treats it as known. To check that our results survive parameter uncertainty, we instead estimate the five model parameters and the four forward guidance durations jointly, using Bayesian methods. We build on Denes and Eggertsson (2009), who form priors over the parameters of a New Keynesian model at the zero lower bound and

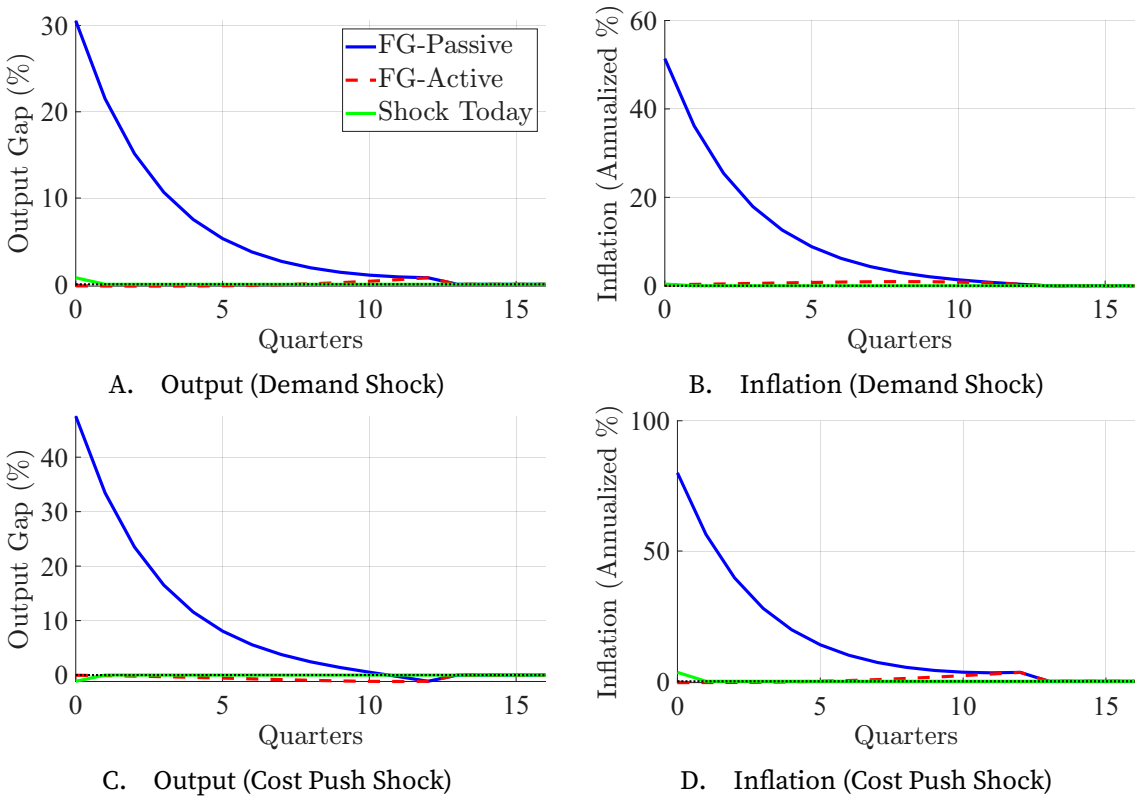



FIGURE A3. Responses to News Shocks: Predetermined Path vs Active Policy Rule

*Note*: Response of output, inflation, and the policy rate to news about future demand or cost push shocks. Blue line assumes a predetermined path for the policy rate ($\phi_\pi = \phi_y = 0$), while the dashed red line assumes an active policy rule satisfying the Taylor principle (we assume $\phi_\pi = 1.5$, $\phi_y = 0.5/4$, but the result does not depend on this assumption as long as the policy rule satisfies condition (1)). The green line presents the effects of the same shock if it occurs today, at time $t = 0$.

use the model to match the trough of the Great Depression exactly. We match the trough and the four regime changes, allowing for error, in the spirit of the Bayesian calibration approach proposed by Eggertsson (2026).

*Priors and likelihood.* We estimate $\theta = (\beta, \alpha, \kappa, \sigma, r_S^n, T_1, T_2, T_3, T_4)$. The priors, listed in Table A1, are centered on the point calibration in section 4. $\beta$ and $\alpha$ have beta priors since they must fall between 0 and 1; $\kappa$, $\sigma$, and $-r_S^n$ have gamma priors since they must be greater than zero; and the durations, which are whole months, have negative binomial priors.

The we target output and year-over-year CPI inflation in the trough (June 1932–March 1933) and in the window following each regime change. For a given $\theta$, Equation 16 and Equation 17 give model output and annualized inflation in each of these five periods. We assumed that each data target equals its model counterpart plus an independent Gaussian error, with a standard deviation of $\omega_y = 4.9$ percentage points for output and $\omega_\pi = 3.0$

TABLE A1. Bayesian Estimation: Prior and Posterior Distributions

| | Prior | | | Posterior | | |
|---|---|---|---|---|---|---|
| Parameter | Distribution | Mean | SD | Median | 5% | 95% |
| A. Model parameters | | | | | | |
| $\beta$ | Beta | 0.9986 | 0.0007 | 0.9984 | 0.9971 | 0.9992 |
| $\alpha$ | Beta | 0.0513 | 0.0154 | 0.0523 | 0.0362 | 0.0732 |
| $\kappa$ | Gamma | 0.0015 | 0.0007 | 0.0010 | 0.0005 | 0.0017 |
| $\sigma$ | Gamma | 1.1613 | 0.5807 | 1.4034 | 0.8838 | 2.1780 |
| $-r_S^n$ | Gamma | 6.1663 | 1.8499 | 6.2243 | 3.9032 | 9.3106 |
| B. Months of forward guidance | | | | | | |
| $T_1$: FDR takes office | Neg. binomial | 14 | 7.0 | 14 | 6 | 24 |
| $T_2$: Eccles nominated | Neg. binomial | 36 | 18.0 | 36 | 24 | 54 |
| $T_3$: Mistake of 1937 | Neg. binomial | 15 | 7.5 | 15 | 6 | 25 |
| $T_4$: Reversal of 1938 | Neg. binomial | 37 | 18.5 | 36 | 24 | 53 |

*Note*: Priors are independent, with means at the point calibration for the model parameters and at the yield-curve durations for $T_i$; $-r_S^n$ is in annualized percent and $T_i$ in months. Draws for which the model has no bounded solution are rejected. The likelihood treats average output and year-over-year inflation in the trough and in each episode's window as observed with independent Gaussian errors, with standard deviations of 4.9 percentage points for output and 3.0 for annualized inflation. The posterior pools 4 random-walk Metropolis chains of 2,000,000 draws each, started from independent prior draws; dropping the first 20% of each chain as burn-in leaves 6,400,000 draws. Acceptance rates are 14%. For every parameter, the rank-normalized split-$\hat{R}$ is at most 1.0001 and the bulk and tail effective sample sizes are at least 68,097.

percentage points for inflation.[36] Indexing the trough by $i = 0$ and the windows following the four regime changes by $i = 1, \ldots, 4$, the measurement equations are

$$y_i^{\text{Data}} = y_i^{\text{Model}}(\theta) + u_i^y, \qquad u_i^y \sim \mathcal{N}(0, \omega_y^2), \tag{A15}$$

$$\pi_i^{\text{Data}} = \pi_i^{\text{Model}}(\theta) + u_i^\pi, \qquad u_i^\pi \sim \mathcal{N}(0, \omega_\pi^2), \tag{A16}$$

where the errors are independent across periods and between output and inflation, and the model values depend on the forward guidance durations only through $T_i$, with $T_0 = 0$ in the trough. The posterior is therefore proportional to the prior times the Gaussian likelihood of the ten targets,

$$p(\theta \mid \text{Data}) \propto p(\theta)\,\mathbf{1}\{\theta \in \Theta\} \times \exp\left\{-\sum_{i=0}^{4}\left[\frac{\left(y_i^{\text{Data}} - y_i^{\text{Model}}(\theta)\right)^2}{2\omega_y^2} + \frac{\left(\pi_i^{\text{Data}} - \pi_i^{\text{Model}}(\theta)\right)^2}{2\omega_\pi^2}\right]\right\}, \tag{A17}$$

[36] These correspond to error variances of 10 and 30 percent of the variance of the output and inflation averages across the five periods. To remain consistent with the point calibration, which primarily targets output to capture the magnitude of the downturn, we keep the output error small so that the output data discipline the estimates.

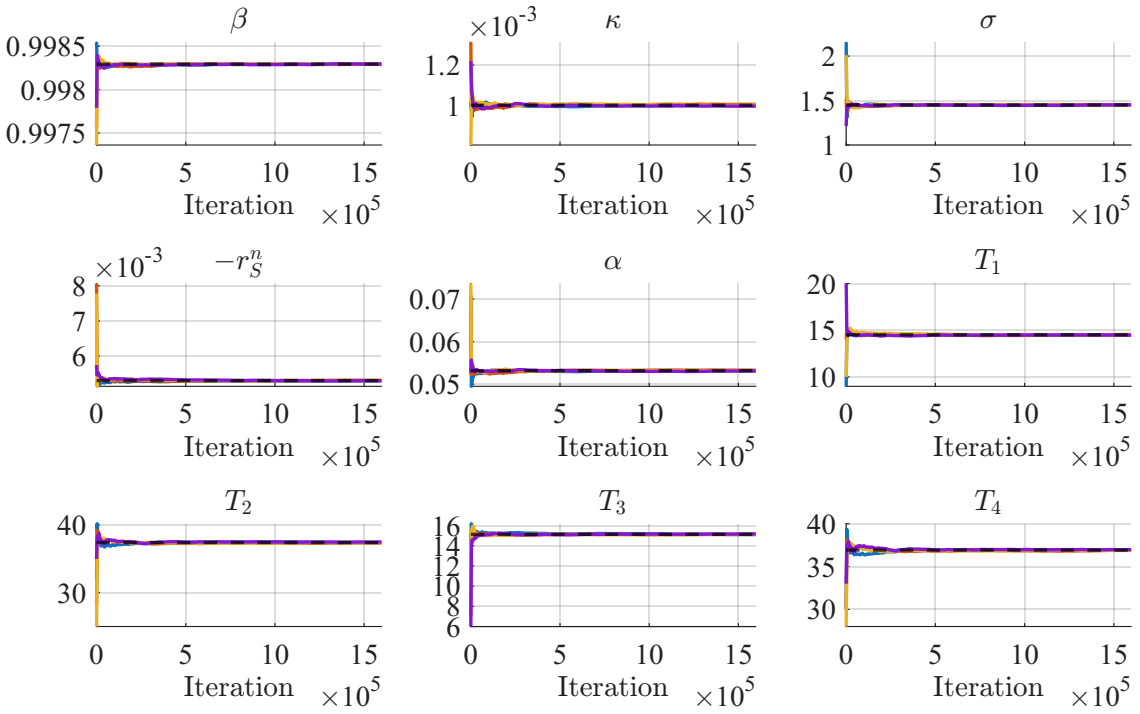



FIGURE A4. Running (Cumulative) Posterior Mean for Each Parameter of the Great Depression Model

*Note*: Each line shows the post-burn-in running posterior mean for one of the four independent chain. For each parameter, the running means converge to the same posterior mean, providing further evidence of convergence.

where $p(\theta) = p(\beta)\,p(\alpha)\,p(\kappa)\,p(\sigma)\,p(-r_S^n)\prod_{i=1}^{4} p(T_i)$ is the product of the independent priors, and $\Theta$ is the set of parameter values that satisfy the condition for a bounded solution, $\alpha(1-\beta(1-\alpha))-\sigma\kappa(1-\alpha) > 0$. Parameter values that violate it have zero posterior probability.

*Sampling.* We sample the posterior with a random-walk Metropolis-Hastings algorithm (An and Schorfheide 2007; Herbst and Schorfheide 2016). To tune the proposal, we find the posterior mode of the five model parameters, holding the durations at their yield-curve values, and run short pilot chains from the mode with the inverse Hessian as the proposal covariance. The covariance of the pilot draws, scaled by $2.38/\sqrt{9}$ (Roberts, Gelman, and Gilks 1997), is then the proposal for four main chains that move all nine parameters jointly, with the proposed steps in the durations rounded to whole months. Although the main chains start from independent draws from the prior, all four converge to the same distribution: the rank-normalized split-$\hat{R}$ of Vehtari et al. (2021) is at most 1.0001 for every parameter, well below their recommended threshold of 1.01. Each chain consists of 2,000,000 draws with a 20 percent burn-in. Figure A4 plots the post-burn-in running

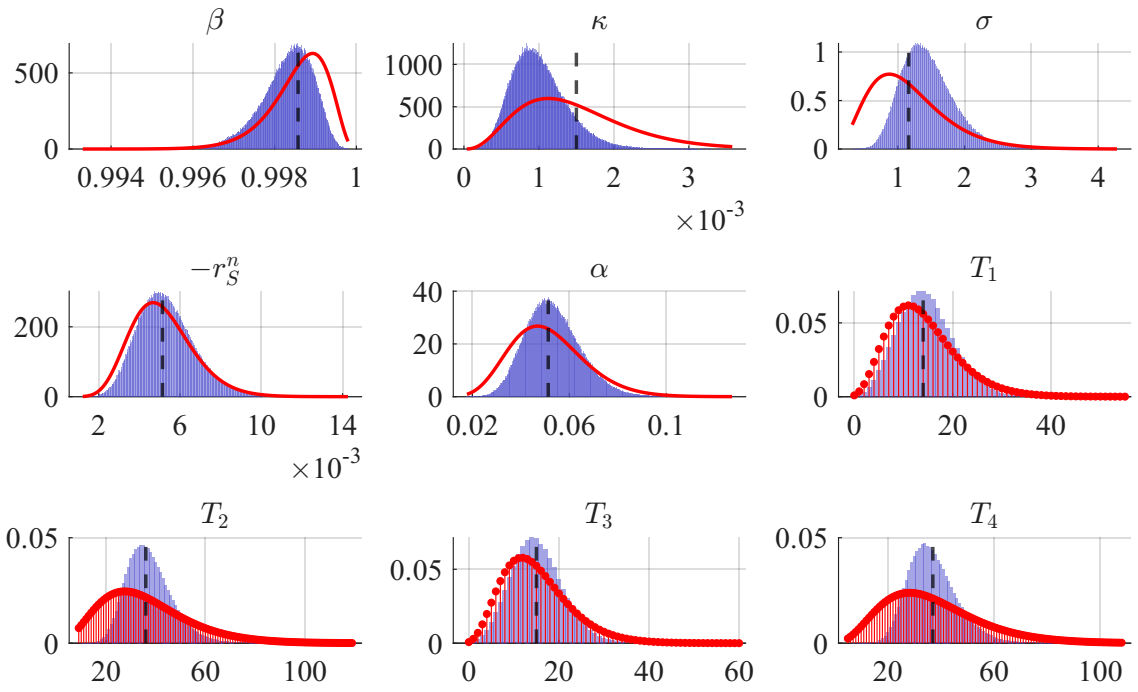



FIGURE A5. Prior and Posterior Distributions of the Great Depression Model

*Note*: Blue histograms show the posterior, pooling the post-burn-in draws of the four chains. Red lines show the prior densities. Dashed black lines mark the prior means: the point calibration for the model parameters and the yield-curve durations for $T_i$. Unlike in Table A1, $-r_S^n$ is a monthly rate in decimal units.

posterior mean for each of the parameters, providing further evidence of convergence.

*Results.* Table A1 summarizes the posterior, and Figure A5 plots the full prior and posterior distributions. The posterior medians of the forward guidance durations are 14, 36, 15, and 36 months, against 14, 36, 15, and 37 months from the yield curve. The posterior probability that forward guidance lengthened after Eccles' nomination, shortened in 1937, and lengthened again in 1938 exceeds 0.999 in each case. The durations chosen to match output in Table 2, 17, 36, 20, and 37 months, all lie within the posterior 90 percent intervals.

The posterior median of $\kappa$ falls from 0.0015 to 0.00096, close to the estimate of Hazell et al. (2022) discussed in subsection 4.2, and that of $\sigma$ rises from 1.16 to 1.40. A flatter Phillips curve brings the model closer to matching the muted inflation response to the reversal of 1938, at the cost of less deflation and a slightly smaller collapse in output during the initial downturn. At the posterior median, output in the trough is 26.1 percent below its July 1929 level and deflation is 5.7 percent, against 30.0 and 10.3 percent in the data (Figure A6).

The estimated model reproduces the swing in output at each regime change. The posterior medians of the change in average output from one window to the next are 7.7, 20.7, −20.4, and 19.9 percentage points, against 10.0, 22.9, −20.4, and 22.9 in the data, and each

observed change lies within its posterior 90 percent interval. Accounting for parameter uncertainty thus leaves our conclusions intact: the output and inflation data are consistent with the forward guidance durations implied by the yield curve, and with these durations the model generates output swings of the size observed in the data.

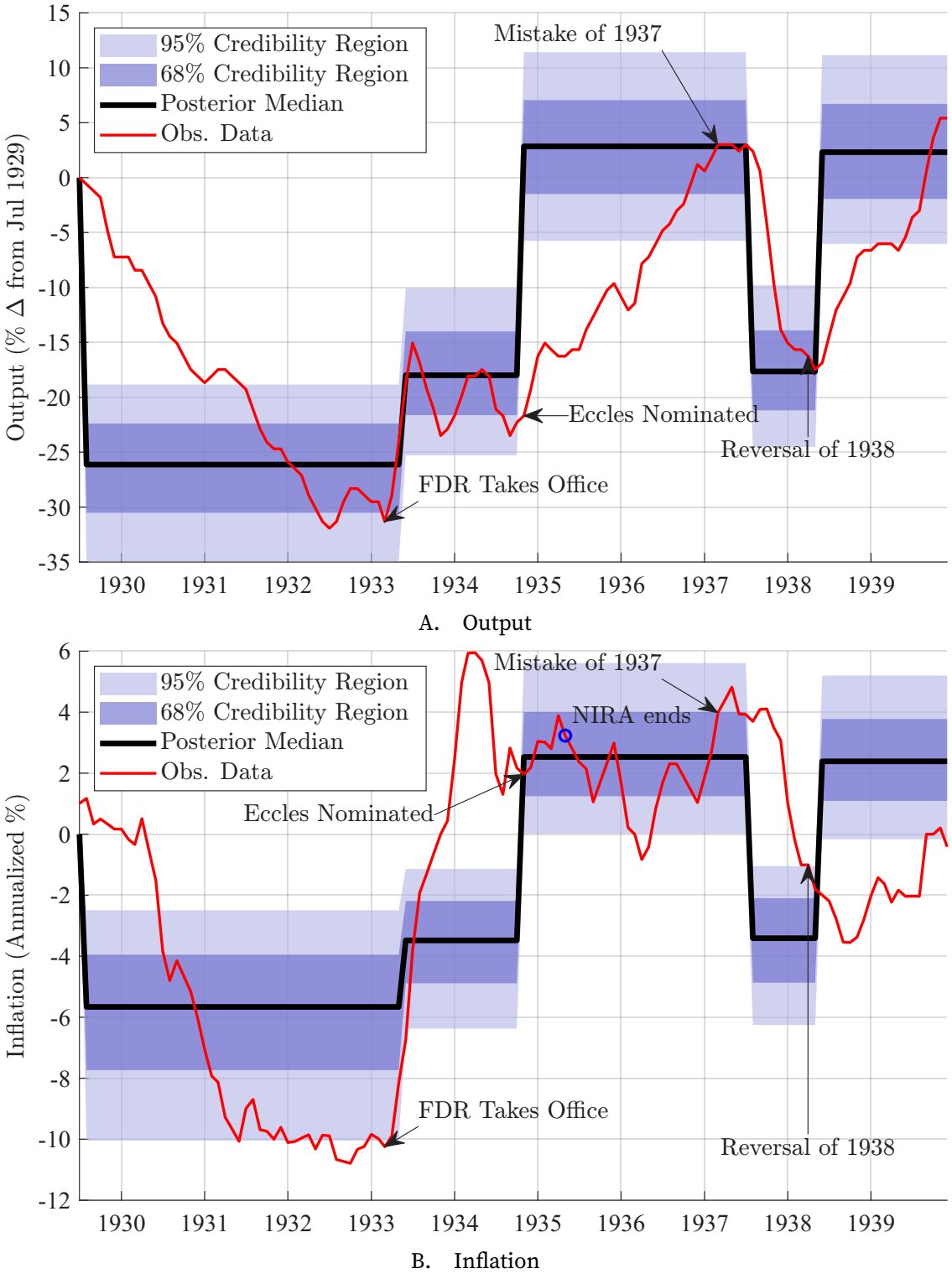



FIGURE A6. Response of Output and Inflation in Great Depression: Estimated Model vs. Data

*Note*: Black lines are the posterior medians of model output (top) and inflation (bottom) in the trough and after each regime change, with each posterior draw using its own forward guidance durations $T_i$. Shaded areas are 68 and 95 percent posterior credible bands for these model values, which exclude the measurement error. Red lines are the data, as in Figure 8.

### C.2. Parameter Sensitivity

The Bayesian estimation in Appendix C.1 lets all nine parameters move together. Here we instead ask how our results depend on the value of individual parameters. We focus on the parameters we calibrate from external evidence rather than from output and inflation: $\alpha$ and the forward guidance durations $T_1$, $T_2$, $T_3$, and $T_4$, which we infer from 3-year yields, and $\beta$ and $r_S^n$, which we take from the natural rate estimates of Schmitt-Grohé and Uribe (2025). The parameters derived from 3-year yields deserve particular scrutiny, because they rest on our term-premium adjustment and on our assumption that changes in term-premium-adjusted yields around each regime change reflect only changes in expectations about future policy.

For each of these parameters, we hold it fixed at three values and re-estimate the remaining eight parameters using the Bayesian approach of Appendix C.1. The three values are the point calibration from subsection 4.2 and the 5th and 95th percentiles of the parameter's prior in Table A1. For $\alpha$, these values imply that the natural rate is expected to remain in its depressed state for 19.5, 34.8, and 12.7 months, respectively. We keep the priors, targets, and likelihood unchanged, so each run samples the posterior of the other eight parameters conditional on the fixed value, a slice of the joint posterior in Equation A17. Each run uses the same three-stage sampler as the joint estimation—a mode search, pilot chains, and four main chains started from prior draws—with 200,000 draws per main chain and a 20 percent burn-in.

Figures A7 through A13 report the results, one figure per fixed parameter. Panel (a) plots each free parameter's conditional posterior median and 90 percent credible interval against the fixed value. For comparison, the gray band shows the median and 90 percent interval from the joint estimation in Appendix C.1, the dashed line marks the point calibration, which is also the prior mean, and the dotted vertical line marks the fixed parameter's calibrated value. In place of a posterior, the fixed parameter's own panel plots the median log-likelihood across posterior draws, which shows which of the three values the output and inflation data favor. Panels (b) and (c) plot model output and inflation at each run's posterior mode against the data, as in Figure 8.

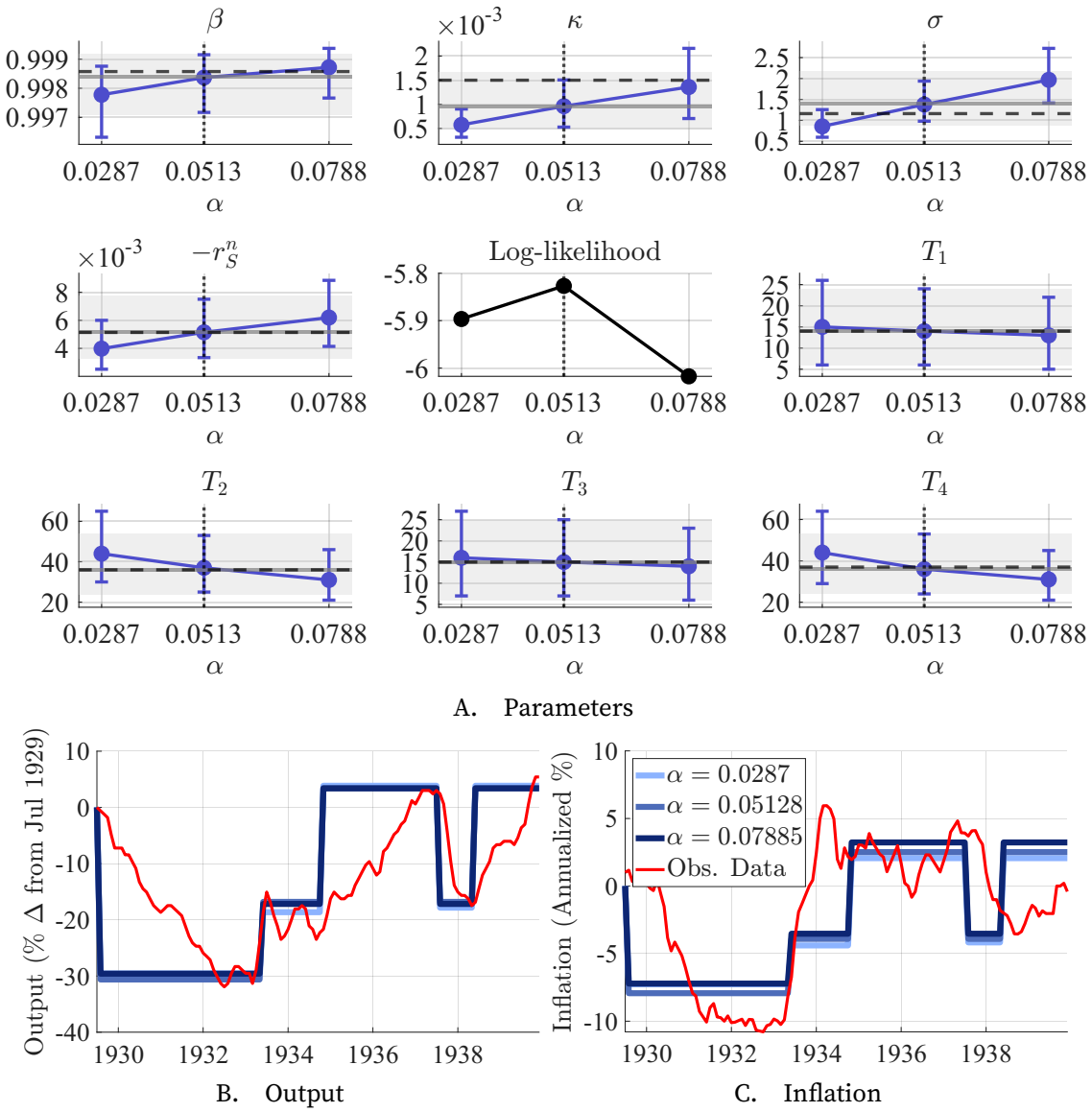



FIGURE A7. Sensitivity of Results to $\alpha$

*Note*: Figure A7A plots the posteriors of the other parameters with $\alpha$ held fixed at the 95th percentile of its prior, the point calibration, and the 5th percentile of its prior, corresponding to expected durations of the depressed state of 12.7, 19.5, and 34.8 months, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $\alpha$. The $\alpha$ panel plots the median log-likelihood across posterior draws. Figure A7B and Figure A7C plot model output and inflation at each run's posterior mode, with darker lines for higher values of $\alpha$. Red lines are the data, as in Figure 8.

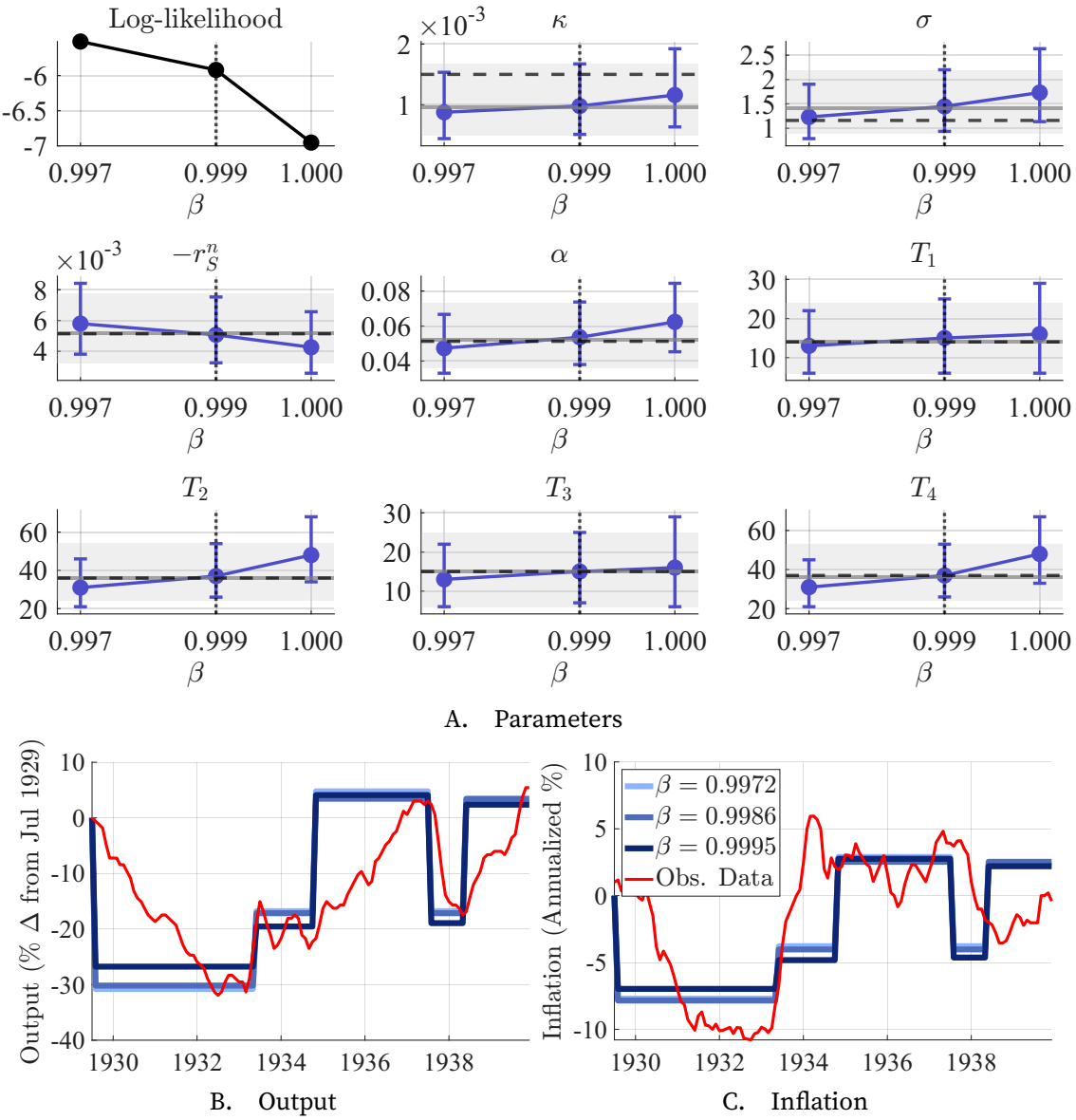



FIGURE A8. Sensitivity of Results to β

*Note*: Figure A8A plots the posteriors of the other parameters with β held fixed at the 5th percentile of its prior, the point calibration, and the 95th percentile of its prior, corresponding to an annualized $r_L^n$ of 3.32, 1.71, and 0.58 percent, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated β. The β panel plots the median log-likelihood across posterior draws. Figure A8B and Figure A8C plot model output and inflation at each run's posterior mode, with darker lines for higher values of β. Red lines are the data, as in Figure 8.

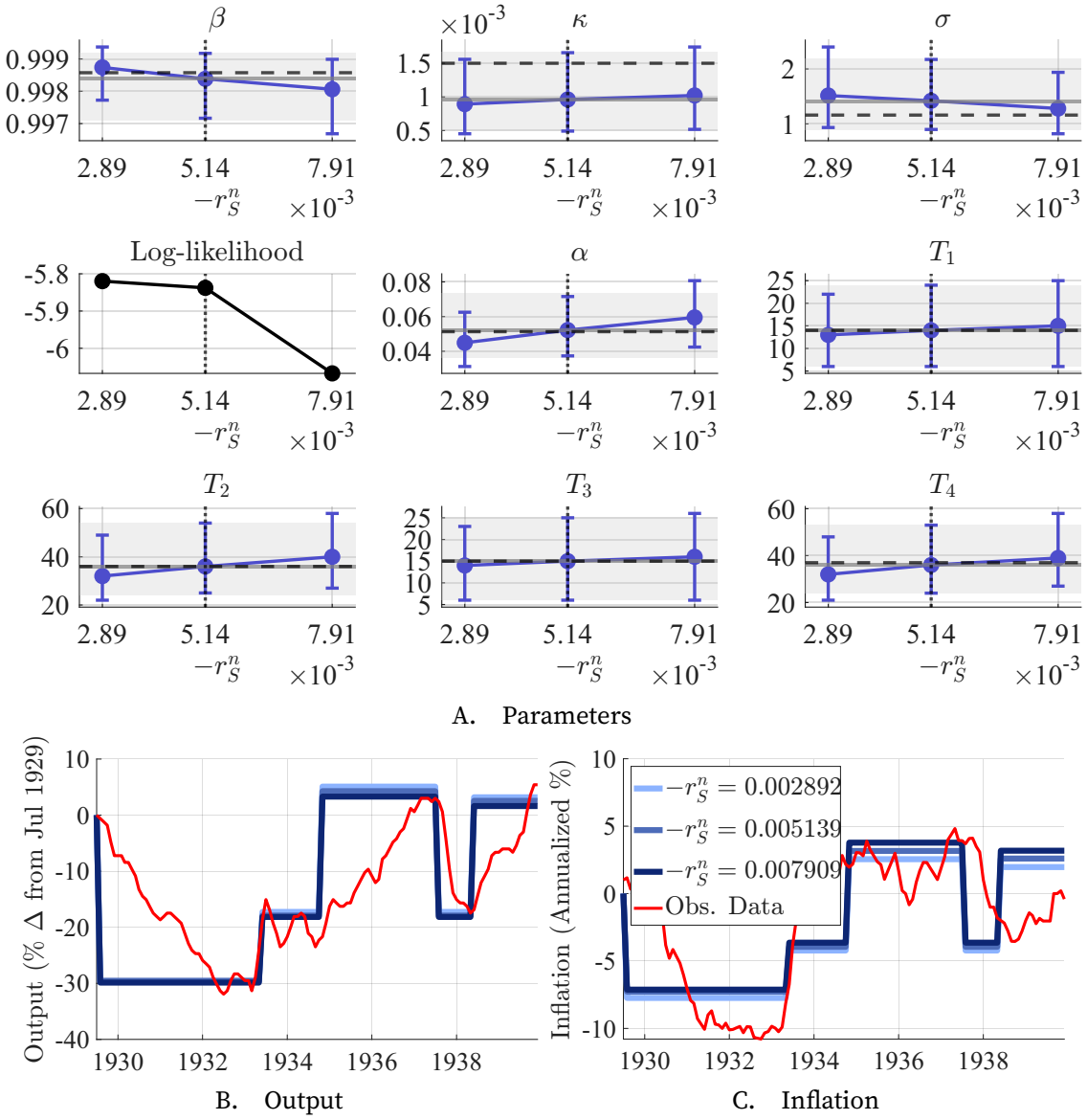



FIGURE A9. Sensitivity of Results to $r_S^n$

*Note*: Figure A9A plots the posteriors of the other parameters with $-r_S^n$ held fixed at the 5th percentile of its prior, the point calibration, and the 95th percentile of its prior, corresponding to an annualized $r_S^n$ of −3.47, −6.17, and −9.49 percent, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $-r_S^n$. The $-r_S^n$ panel plots the median log-likelihood across posterior draws. Unlike in Table A1, $-r_S^n$ is a monthly rate in decimal units. Figure A9B and Figure A9C plot model output and inflation at each run's posterior mode, with darker lines for larger values of $-r_S^n$. Red lines are the data, as in Figure 8.

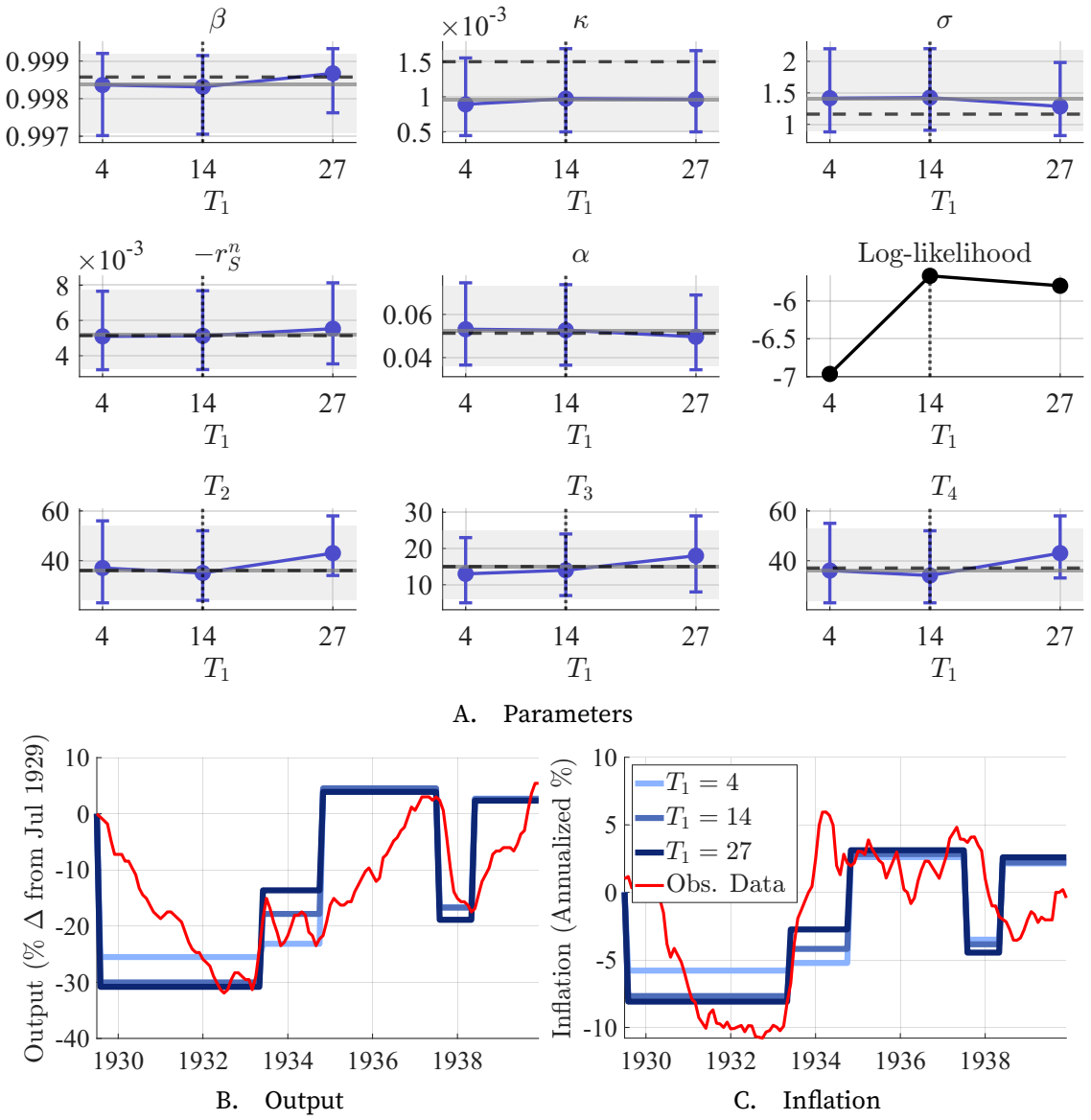



FIGURE A10. Sensitivity of Results to $T_1$

*Note*: Figure A10A plots the posteriors of the other parameters with $T_1$ held fixed at the 5th percentile of its prior, the yield-curve calibration, and the 95th percentile of its prior: 4, 14, and 27 months, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $T_1$. The $T_1$ panel plots the median log-likelihood across posterior draws. Figure A10B and Figure A10C plot model output and inflation at each run's posterior mode, with darker lines for higher values of $T_1$. Red lines are the data, as in Figure 8.

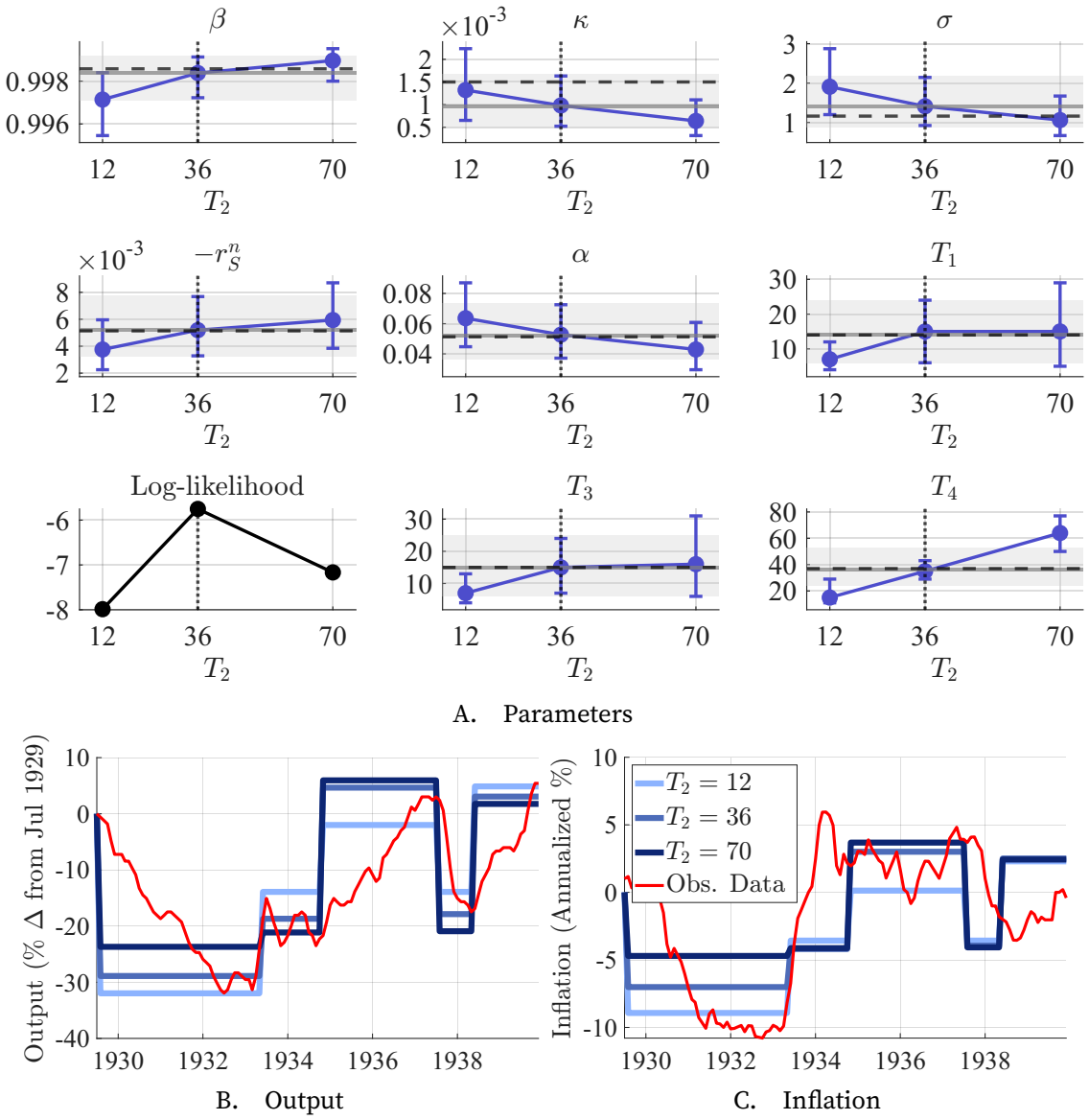



FIGURE A11. Sensitivity of Results to $T_2$

*Note*: Figure A11A plots the posteriors of the other parameters with $T_2$ held fixed at the 5th percentile of its prior, the yield-curve calibration, and the 95th percentile of its prior: 12, 36, and 70 months, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $T_2$. The $T_2$ panel plots the median log-likelihood across posterior draws. Figure A11B and Figure A11C plot model output and inflation at each run's posterior mode, with darker lines for higher values of $T_2$. Red lines are the data, as in Figure 8.

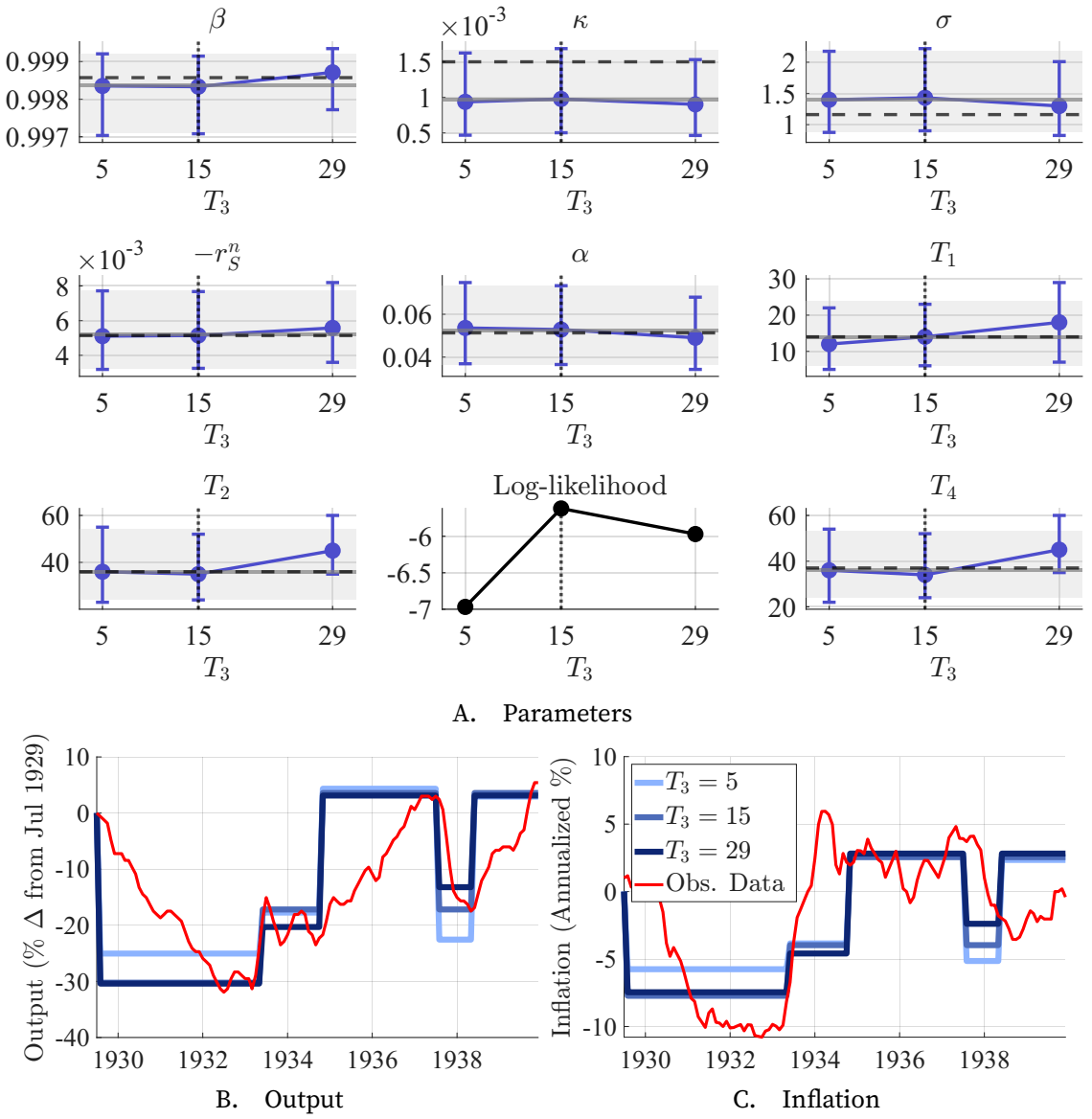



FIGURE A12. Sensitivity of Results to $T_3$

*Note*: Figure A12A plots the posteriors of the other parameters with $T_3$ held fixed at the 5th percentile of its prior, the yield-curve calibration, and the 95th percentile of its prior: 5, 15, and 29 months, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $T_3$. The $T_3$ panel plots the median log-likelihood across posterior draws. Figure A12B and Figure A12C plot model output and inflation at each run's posterior mode, with darker lines for higher values of $T_3$. Red lines are the data, as in Figure 8.

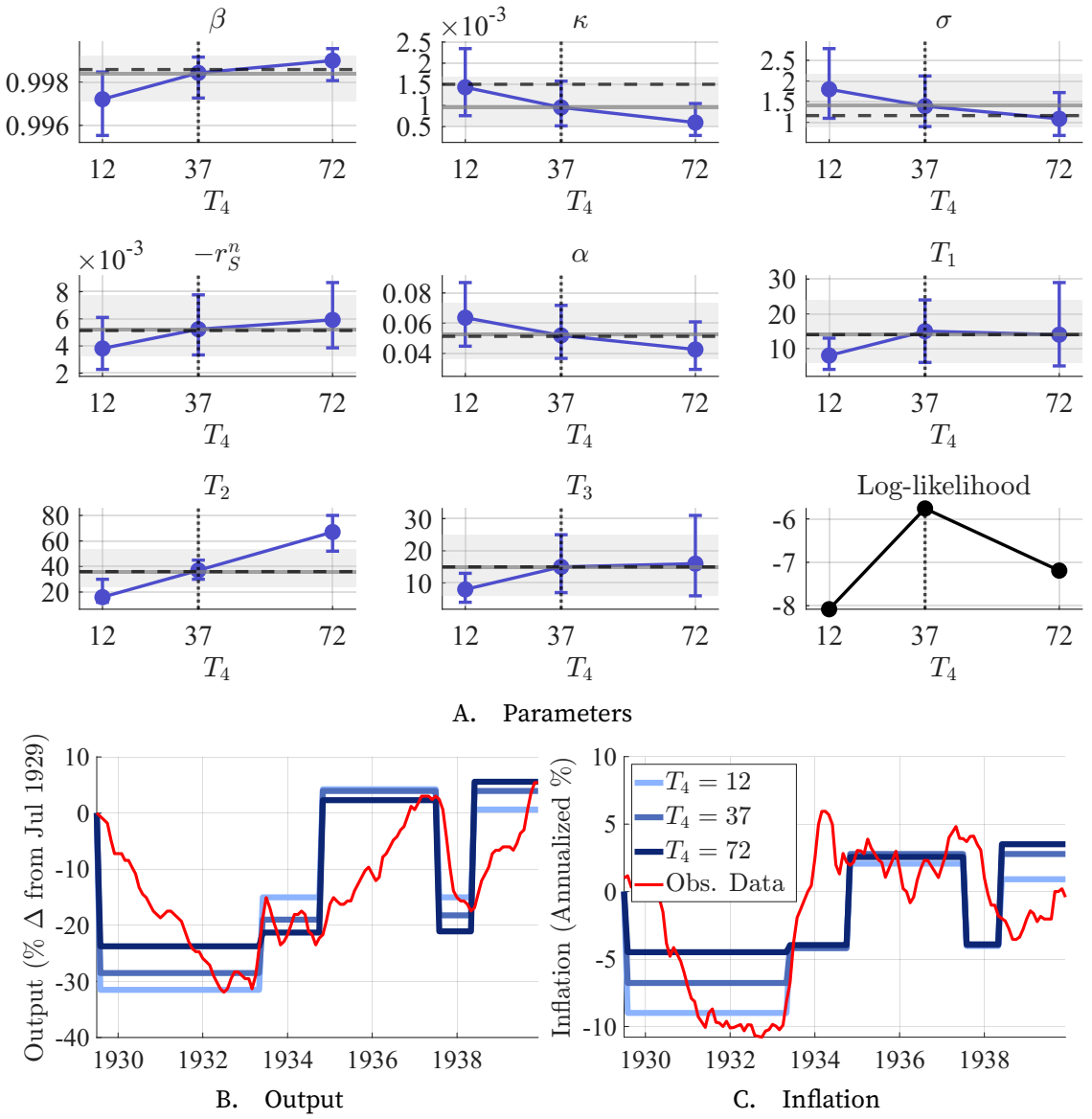



FIGURE A13. Sensitivity of Results to $T_4$

*Note*: Figure A13A plots the posteriors of the other parameters with $T_4$ held fixed at the 5th percentile of its prior, the yield-curve calibration, and the 95th percentile of its prior: 12, 37, and 72 months, respectively. Blue points are posterior medians and bars are 90 percent credible intervals. The gray line and band are the posterior median and 90 percent credible interval from the joint estimation in Appendix C.1. Dashed lines mark the point calibration, and the dotted vertical line marks the calibrated $T_4$. The $T_4$ panel plots the median log-likelihood across posterior draws. Figure A13B and Figure A13C plot model output and inflation at each run's posterior mode, with darker lines for higher values of $T_4$. Red lines are the data, as in Figure 8.

TABLE A2. Forward Guidance by Episode with Habit Formation: Yield Curve vs. Model

| Episode | 3-year yield change (bp) | Months of FG, $T_i$ | | Output (% from July 1929) | | |
|---|---|---|---|---|---|---|
| | | Yield curve | Matched | Yield curve | Matched | Data |
| 1. FDR takes office | −52 | 14 | 20 | −24.3 | −20.4 | −20.0 |
| 2. Eccles nominated | −44 | 36 | 37 | 0.9 | 2.9 | 2.9 |
| 3. Mistake of 1937 | 40 | 15 | 0 | −11.7 | −15.6 | −17.5 |
| 4. Reversal of 1938 | −45 | 37 | 40 | 1.2 | 4.9 | 5.4 |

*Note*: $T_i$ is the number of months the policy rate stays at zero after the natural rate returns to $r_L^n$, under the forward guidance (FG) in place after episode $i$; yield changes and yield-curve durations are as in Table 2. With habits, output adjusts gradually. Without FG, model output falls to −30.0% by March 1933, the end of the trough; $T_i$ then applies over April 1933–November 1934, December 1934–July 1937, August 1937–May 1938, and June 1938–December 1939, and model output is its level at the end of each. Matched durations bring that level closest to average output in the data over June 1933–November 1934, March–July 1937, May 1938, and November–December 1939 when output starts each episode at its average in the data over the previous window, or over the trough for episode 1. Output in the data is industrial production relative to July 1929, rescaled to average −30.0% over the trough, June 1932–March 1933.

### C.3. Habit Formation

In the model from section 4, output immediately jumps in response to any changes in monetary policy, whereas output in the data moves more gradually. This section demonstrates that allowing for habit formation, which in turn allows us to match the more gradual changes in output observed in the data, does not qualitatively change our conclusions. Realistically sized forward guidance shocks can still lead to large changes in output, but these changes in output are consistent with the changes we observe in the data.

The form of habit formation we choose is simple enough to maintain the structure of the basic New Keynesian model, with a straightforward reinterpretation of variables as quasi-growth rates. With the simple form of habit formation the IS and PC become

$$\tilde{y}_t = \mathbb{E}_t \tilde{y}_{t+1} - \sigma(i_t - \mathbb{E}_t \pi_{t+1} - r_t^n) \tag{A18}$$

$$\pi_t = \kappa \tilde{y}_t + \beta \mathbb{E}_t \pi_{t+1} \tag{A19}$$

Where $\tilde{y}_t = y_t - \gamma y_{t-1}$ is the quasi-growth rate of output. Eggertsson (2008) provides detailed derivations.

We set the additional parameter $\gamma$ to 0.92, the monthly equivalent of the estimates for habit persistence in Gelain and Lopez (2023). We then recalibrate the model using the same procedures described in subsection 4.2. We report the resulting calibrated parameter values in Table A3. The yield based parameters remain unchanged, as they are calibrated on the same data and with the same methods as the parameters in section subsection 4.2. Since the model equations now pin down quasi-output growth rather than

TABLE A3. Calibration of the Great Depression Model with Habit Formation

| Parameter | | Value | Calibration target |
|---|---|---|---|
| $\beta$ | Discount factor | 0.9986 | $r_L^n = -\log\beta$ |
| $\alpha$ | Probability that $r_t^n$ reverts to $r_L^n$ | 0.0513 | 3-year yield, end of February 1933 |
| $\kappa$ | Slope of the Phillips curve | 0.0192 | Inflation/output in the trough |
| $\sigma$ | Intertemporal elasticity of substitution | 0.0909 | Output in the trough |
| $\gamma$ | Habit persistence | 0.9242 | Gelain and Lopez (2023) |
| $r_S^n$ | Natural rate in the depressed state | −6.1663 | Natural rate, 1933 |
| $r_L^n$ | Long-run natural rate | 1.7132 | Natural rate, 1941–2000 |

*Note*: The model is monthly; natural rates are annualized percentages. $\beta$, $\alpha$, $r_S^n$, and $r_L^n$ are calibrated as in Table 3. With habit formation, output each month is $\gamma$ times its level in the previous month plus output in the model of subsection 4.1, so it adjusts gradually. $\gamma$ is set so that $\gamma^3$ equals 0.7893, the quarterly estimate of habit persistence in Gelain and Lopez (2023). $\kappa$ and $\sigma$ are then set so that, without forward guidance, the model matches the averages over the trough, June 1932–March 1933, used in Table 3: −10.3% for year-over-year CPI inflation and −30.0% for output. Model output falls gradually from its July 1929 level and reaches −30.0% in March 1933, 44 months later.

output, we adjust our calibration for $\kappa$ to reflect this.

$$\kappa = \frac{\pi}{\tilde{y}}(1-\beta(1-\alpha)) = \frac{\pi}{y}\left[\frac{1-\gamma^D}{1-\gamma}\right](1-\beta(1-\alpha))$$

Where $D$ is the number of months from July 1929 to the trough in February 1933. This gives a value of $\kappa = 0.0193$. We can then recalibrate $\sigma$ to match the size of the downturn, giving $\sigma = 0.0900$.

As figure A14 demonstrates, our conclusions still holds when we account for habit formation to match the gradual changes in output. The duration of forward guidance suggested by 3-year yields leads to paths for both output and inflation that are broadly in line with the data. Similarly, choosing forward guidance to match the path of output leads to forward guidance durations that are close to those suggested by yields.

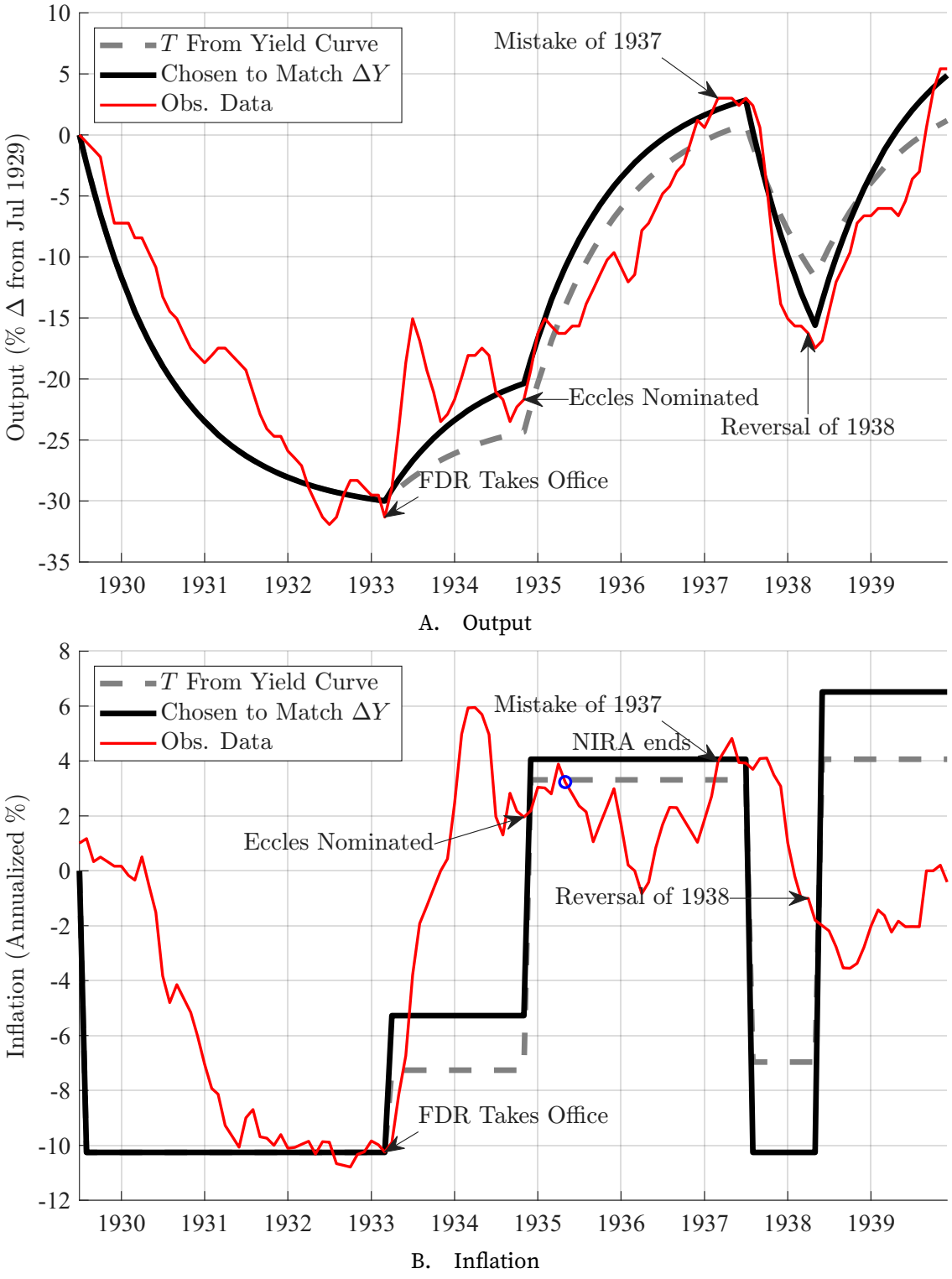



FIGURE A14. Response of Output and Inflation in Great Depression: Model Vs. Data with Habit Formation

*Note*:Solid black lines are response of output (top) and inflation (bottom) to forward guidance picked to match output following regime changes. Dashed gray lines are are response of output (top) and inflation (bottom) to the amount of forward guidance suggested by the yield curve.